

1 Carryover representation and inference for biomarker-treatment
2 interaction tests in aggregated N-of-1 trials under covariance
3 moderation: a factorial simulation comparison of nine analysis
4 specifications

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41 Abstract

42 **Background.** Aggregated N-of-1 trials pool repeated within-patient on/off contrasts to vali-
43 date predictive biomarkers. Under covariance moderation, a multivariate-normal differential-
44 correlation architecture in which the biomarker-treatment interaction lives in a treatment-
45 state-dependent correlation, a companion study found that carryover costs up to roughly
46 31% of power in relative terms. The analyst must still decide how to represent residual expo-
47 sure in the linear predictor. We do not yet know which representation is most robust to the
48 ordinarily unknown pharmacokinetic decay shape.

49 **Methods.** We compare nine analysis-model specifications for the drug-exposure regressor,
50 in two tiers. The Tier 1 comparison, our primary evidence base, covers three specifications
51 across a full factorial grid. Unadjusted is the binary on-drug indicator, ignoring carryover
52 altogether. Lag-adjusted augments that indicator with a lagged just-off-drug term that es-
53 timates carryover non-parametrically, the classical crossover device. Exposure-weighted is a
54 continuous indicator that commits to a parametric decay and half-life, the current package
55 default. We embed these three in an ADEMP-structured factorial simulation¹, crossing two
56 data-generating decay forms (exponential, Weibull) with three trial designs, two sample sizes,
57 and three interaction strengths (216 cells, 500 replicates each), plus sensitivity blocks and a
58 cluster-robust recalibration. We fit all three to the same simulated datasets, so every con-
59 trast is paired, and we report each performance measure with its Monte Carlo standard error

(MCSE). A Tier 2 extension adds six further specifications at a single reference cell: an AIC-selected half-life variant, a paired-difference regression with no repeated-measures model at all, and CR2 cluster-robust variants of four of the above. This second tier is a more targeted comparison, not a full grid.

Results. Three findings stand out. First, the lagged-treatment term buys nothing. Lag-adjusted and Unadjusted are statistically indistinguishable across all 216 Tier 1 cells. They differ by a mean of 0.003 in power against an MCSE of 0.018, and agree to three decimal places on bias, mean squared error, and coverage. The two share an estimand and differ by exactly one nuisance term, so this is a clean null. Second, Exposure-weighted attains the highest Tier 1 power only in the two designs with a blinded-discontinuation phase (Hybrid and OL+BDC), where its advantage is about 10 percentage points (0.860 vs 0.766 for Unadjusted at the reference cell). Under the classical crossover (CO) design it is markedly inferior (0.488 vs 0.830). Where it leads, the lead is robust to half-life mis-specification and to autocorrelation strength. It widens under a light-tailed decay-shape mis-specification but narrows to statistical insignificance under the two most heavy-tailed Weibull shapes we examined. The model-based interaction test is mildly conservative for all three (mean null rejection 0.036 to 0.039), an artifact of the working covariance that a CR2 cluster-robust standard error removes without disturbing the ranking. Third, at the Hybrid reference cell the Tier 2 extension shows that combining the AIC-selected half-life with a CR2 standard error outperforms every other specification we tested, including the CR2-corrected Exposure-weighted predictor, with Type I error held at nominal despite the post-selection risk that the half-life selection step raises in principle. This result held at only that one cell, so we report it as a promising direction rather than a validated recommendation.

Conclusions. Four points warrant emphasis. First, under the covariance-moderation architecture we recommend the exposure-weighted predictor, reported with a cluster-robust standard error, for the Hybrid and OL+BDC designs. Under a crossover design it should not be used; the unadjusted binary indicator is materially better there. Second, we cannot recommend the classical lagged-treatment remedy in any design examined here, since it never improved on ignoring carryover altogether. We trace this to its entering the model as a nuisance main effect, which leaves the attenuation of the interaction untouched. Third, across all conditions anticipated dropout governs power far more than the choice among specifications, and design effort should be weighted accordingly. Fourth, the AIC-selected half-life specification with a CR2 correction merits further validation as a candidate default. It outperformed every other specification at the one cell where we compared all nine directly, but we have not yet tested it against the same robustness battery the Tier 1 three have. This manuscript restricts the grid to the covariance-moderation architecture and to the exponential and Weibull decay forms. We evaluate the mean-moderation architecture and linear decay in the archived companion drafts `archive/report.Rmd` and `archive/report-slim.Rmd`.

1 Introduction

We begin with the basic design. In a single N-of-1 trial, one patient is crossed repeatedly between active treatment and a comparator. The within-patient contrast between on-drug and off-drug periods estimates that patient’s individual treatment effect. An aggregated N-of-1 trial pools many such series, which allows for population-level inference at sample sizes smaller than a parallel-group trial would require². The estimand is the biomarker-treatment interaction, that is, whether a baseline covariate B_i modifies the treatment effect. We evaluate nine analysis approaches for inference regarding this estimand in a ‘hybrid’ N-of-1 design, defined in Section 2.2.

The main inferential complication in N-of-1 trials is carryover. A drug’s pharmacological effect does not switch off abruptly at discontinuation. It continues to ‘decay’ as a function of the drug’s half-life, so post-discontinuation observations remain somewhat indicative of treatment effect. An exposure regressor that ignores this is, in effect, a mis-measured version of what the patient actually received, and the damage is twofold. First, the interaction contrast is diluted. Occasions still carrying a substantial fraction of the drug effect are coded identically to occasions carrying none, which biases the biomarker-by-exposure coefficient toward zero, as any covariate measured with error would. Second, those same occasions are heterogeneous in a way the model does not represent, so the unexplained variation inflates the standard error of the very coefficient that has just been attenuated. The test loses on both counts, and power falls accordingly.

We note that this is a loss of power, not of validity. Under the null, $c_{bm} = 0$, no biomarker-response association exists at any level of residual exposure, so a mis-coded regressor has no interaction to distort, and test size is unaffected by exposure coding. The mild conservatism observed in Section 3.6 has a separate origin in the working covariance and applies equally across specifications. Carryover mis-specification is therefore a question of efficiency, and we frame the study in terms of recovered power.

Three answers are available for how to represent residual exposure in the linear predictor. These correspond to three positions an analyst can take toward the unknown decay. The analyst may deny it, coding exposure strictly on or off. The analyst may estimate it, adding a nuisance term whose magnitude the data determine. Or the analyst may assume it, committing to a decay curve and a half-life. A companion study³, building on Hendrickson², shows that under the covariance-moderation architecture carryover erodes power directly and design-dependently. In that architecture the interaction lives in the treatment-state-dependent covariance $\text{Cor}(B_i, BR_{it}) = c_{bm} \cdot \phi(t_{sd})$. At a half-life of one week, the relative loss is roughly 31% under OL+BDC but only 3 to 5% under crossover and Hybrid, against 1 to 3% under an alternative architecture where the biomarker moderates the mean response directly. We restrict attention here to covariance moderation, where the carryover problem is largest and where Hendrickson’s original results were obtained. That earlier work fixed the decay shape at exponential and evaluated only one analysis specification, leaving open which representation of residual exposure performs best when the true decay shape is not known precisely, as is ordinarily the case.

We compare three specifications as our primary evidence base. Unadjusted is the binary on-drug indicator, ignoring carryover entirely and serving as the benchmark any remedy must beat. Lag-adjusted augments it with a lagged just-off-drug term that estimates carryover magnitude without assuming a functional form, following the classical crossover-trial device of Jones and Kenward⁴. Exposure-weighted is a continuous indicator committing to a parametric decay form and half-life, the current pmsimstats-ng default. Unadjusted and Lag-adjusted estimate the same coefficient and are fitted to the same simulated datasets, differing by exactly one nuisance term, so their contrast isolates that term's contribution alone. Exposure-weighted differs on a second axis as well, since it changes the regressor carrying the interaction. We cross all three against exponential and Weibull data-generating decay forms and ask whether one can be recommended outright, or whether the choice instead depends on trial design. We follow the ADEMP reporting framework of Morris, White, and Crowther¹ throughout. Six further specifications, an AIC-selected half-life variant, a paired-difference regression, and CR2 cluster-robust variants of four base specifications, are added later at a single reference cell (Section 2.5), once the results from these first three raise specific new questions the Tier 1 grid alone cannot answer. Section 2 describes the simulation design in ADEMP order, Section 3 gives the results, and Section 4 takes up the practical implications.

2 Simulation design

The study simulates complete aggregated N-of-1 trials and analyzes each one under every specification. A single replicate proceeds as follows. Patients are allocated across the randomization paths of the chosen trial design (Section 2.2). For each patient we draw a pre-treatment biomarker together with the three latent response components, namely the pharmacological response, the natural-history trend, and the placebo-belief component. These are drawn jointly across all eight measurement occasions from a multivariate normal distribution whose means follow modified Gompertz trajectories. The covariance carries within-component AR(1) autocorrelation, cross-component correlations, and the treatment-state-dependent biomarker-response correlation that encodes the interaction (Section 2.3). The observed symptom score at each occasion is the patient's baseline severity less the sum of the three components, so that a beneficial effect lowers the score. Carryover enters through a decay function of time since discontinuation, which keeps part of the pharmacological effect present at nominally off-drug occasions. All three analysis specifications (Section 2.4) are then fitted to that same simulated dataset, and we record the interaction coefficient, its standard error, and its p -value from each. This is repeated 500 times for every parameter combination in the grid of Section 2.7.

Fitting all specifications to a common dataset is deliberate. It makes every contrast among them paired, so that differences are estimated with substantially less Monte Carlo variance than independent draws would give. Throughout, N denotes the **total** number of patients in a trial, allocated as evenly as possible across the design's randomization paths, so a design with P paths assigns about N/P patients to each. At $N = 70$ this is 35 per path under the two-path designs and 17 or 18 per path under the four-path Hybrid design.

2.1 Aims

The aim is to quantify, under covariance moderation, the cost of mis-specifying the carry-over representation in the analysis model. A related aim is to determine which of the three candidate strategies performs better across the grid of data-generating decay forms and trial designs. A second aim, addressed in the Tier 2 sensitivity blocks of Section 3.7, is to check whether the resulting ranking survives perturbation of within-subject autocorrelation, dropout rate, half-life accuracy, and interaction size.

2.2 Trial designs

The three designs differ in how on-drug and off-drug occasions are arranged, and so in how much post-discontinuation information each generates. All three schedule eight measurement occasions per patient across randomization paths, each a fixed on-drug/off-drug schedule. Each design also carries an expectancy weight, one during open-label periods and one half during blinded periods, scaling the placebo-belief component.

Crossover (CO). A traditional two-period within-patient crossover, blinded throughout, occasions spaced 2.5 weeks apart from 2.5 to 20 weeks. One path is active for the first four occasions then comparator, and the other reverses that order and so never discontinues, contributing no post-discontinuation observations.

Open-label followed by blinded discontinuation (OL+BDC). Open-label titration at 4, 8, 12, and 16 weeks, then weekly follow-up to week 20 with a blinded late switch to placebo. The two paths discontinue after week 17 or 18, so only two or three occasions follow.

Hybrid. The Hendrickson design and this program's reference. Open-label titration at weeks 4 and 8 is followed by blinded discontinuation weekly from week 9 to 12, then a brief two-occasion crossover at weeks 16 and 20. The four paths vary discontinuation timing and crossover-arm order. It is the richest of the three in on-off transitions and the only one allowing more than one discontinuation per patient.

One consequence bears directly on the results. The first post-discontinuation occasion falls one week after switching under OL+BDC and Hybrid, but 2.5 weeks under CO. At the half-lives examined here, roughly half the pharmacological effect therefore persists into the first off-drug occasion under the two non-crossover designs, against less than a fifth under CO. The designs therefore differ substantially in how much carryover contamination they can accumulate at all (Section 4.2).

2.3 Data-generating mechanism

Under covariance moderation, the biomarker and the biological response are drawn jointly from a multivariate normal distribution with a treatment-state-dependent correlation, $\text{Cor}(B_i, BR_{it}) = c_{bm} \cdot \phi(t_{sd})$, where ϕ is the decay form and t_{sd} the time since drug discontinuation. Mean moderation and the combined dual-channel architecture fall outside this manuscript's scope; the full cross-architecture comparison is in the archived full-scope drafts

(`archive/report.Rmd`, `archive/report-slim.Rmd`). The decay function $\phi : \mathbb{R}_+ \rightarrow [0, 1]$ maps t_{sd} to a residual-effect multiplier. We evaluate two forms here, each matched to a common half-life $t_{1/2}$. A third, linear decay, appears in the full-scope manuscripts but not here.

2.3.1 Exponential decay

$$\phi_{\text{exp}}(t_{sd}) = \exp(-\lambda t_{sd}), \quad \lambda = \frac{\ln 2}{t_{1/2}}$$

This is the decay form used throughout the companion manuscript and implemented in `functions.R` under `implementations/tidyverse/`. It corresponds to classical first-order elimination kinetics, the default pharmacokinetic assumption, and it serves as the reference against which Weibull is compared.

2.3.2 Weibull decay

$$\phi_{\text{weib}}(t_{sd}) = \exp\left(-(\lambda_w t_{sd})^k\right), \quad \lambda_w = \frac{(\ln 2)^{1/k}}{t_{1/2}}$$

A shape parameter k governs whether the elimination tail is heavier than exponential ($k < 1$, slower elimination) or lighter ($k > 1$, faster clearance). It recovers the exponential form exactly at $k = 1$. We report four values, $k \in \{0.25, 0.5, 2.0, 4.0\}$, two on each side of exponential: $k = 0.5$ and the more extreme $k = 0.25$ (progressively more prolonged low-concentration tails), and $k = 2.0$ and the more extreme $k = 4.0$ (progressively faster clearance once concentration falls below some threshold). This is a wider sensitivity range than an earlier version of this grid used ($k = 0.7, 1.5$ only). We note that $k = 1.0$ is not a fifth Weibull condition: it is numerically indistinguishable from Exponential and so is not reported separately, giving five decay-shape levels in the grid rather than six.

Figure 1 makes the shape difference concrete. The $k = 0.25$ and $k = 0.5$ curves have progressively heavier tails, and ϕ stays well above the exponential curve for every $t_{sd} > t_{1/2}$, most markedly at $k = 0.25$. In practical terms, the analysis-side predictor picks up substantially more residual carryover during a prolonged washout than the half-life alone would suggest, and increasingly so as k falls further below one. The $k = 2.0$ and $k = 4.0$ curves, by contrast, have progressively lighter tails and clear sharply faster once concentration falls below the half-maximum threshold. We note that all five curves pass through the same half-life-defining point ($\phi = 0.5$ at $t_{sd} = 1.0$), so the five forms remain comparable at a fixed nominal half-life even though their shapes diverge substantially on either side of it.

2.4 Estimand and analysis specifications

The interaction estimand is the on-drug biomarker-response correlation c_{bm} , a dimensionless quantity. Bias and empirical SE are reported on a calibrated effect-size scale, namely

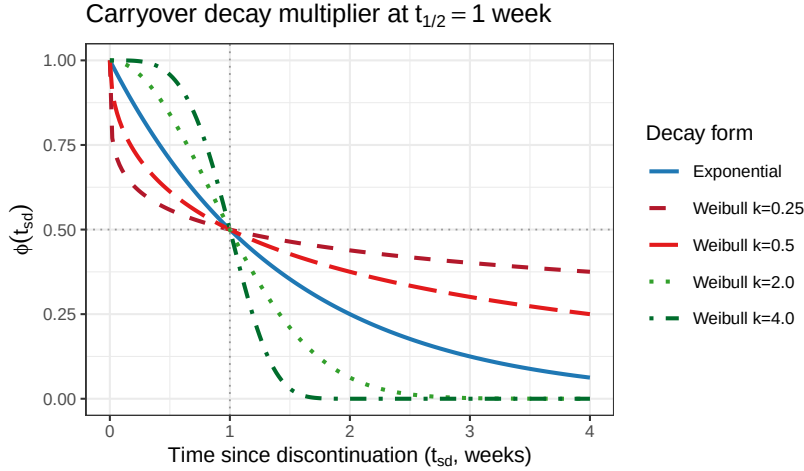


Figure 1: The carryover decay multiplier $\phi(t_{sd})$ at the reference half-life ($t_{1/2} = 1.0$ week) for exponential decay and the four Weibull shapes. The dotted reference lines mark the defining half-life point ($\phi = 0.5$ at $t_{sd} = t_{1/2}$), which all five curves pass through by construction; the shape parameter governs only how quickly ϕ falls before that point and how slowly it decays after.

the expected difference in on-drug response per SD of the biomarker at $t_{1/2} = 0$, per docs/19-mean-moderation-implementation-notes.tex. Power and Type I error are on the native test-statistic scale, with the null estimand $c_{bm} = 0$. The analysis model is `nlme::lme(Sx ~ bm * X + t, random = ~1 | ptID, correlation = corCAR1(form = ~ t | ptID))` for the Tier 1 three. These differ only in how X is constructed and whether a nuisance term is added. Unadjusted and Lag-adjusted both test the coefficient on $bm:D_{it}$, so every performance measure is comparable between them. Exposure-weighted tests $bm:D_{bc,it}$, a different quantity. Power and Type I error remain comparable across all three, since they are properties of the test, but bias, MSE, and coverage are not comparable across that divide (Section 3.1). Nine specifications are evaluated in total across this manuscript: the Tier 1 three below, plus six Tier 2 additions introduced in the ‘Choice of analysis specifications’ section that follows and tested in Blocks S7-S10 (Section 3.7). Internally, and in the code and data files described in the Reproducibility section, each specification is stored under an E-prefixed code (E1, E2, ...). This manuscript’s prose refers to all nine uniformly by the compact G1-G9 code defined in `spec-labels.R`, or by name, per Table 1 below.

2.4.1 Unadjusted (G1): binary on-drug indicator

Here $X = D_{it}$, the binary on-drug indicator (one on drug, zero off). Carryover is not represented at all. An off-drug occasion still carrying most of the pharmacological effect is therefore coded identically to one carrying none. This is the default an analyst obtains without considering carryover, and we treat it as the benchmark the other specifications must justify their additional structure against. It consumes no half-life, so it is invariant to the analyst’s pharmacokinetic assumptions.

Table 1: All nine analysis specifications evaluated in this manuscript. G1-G3 are the Tier 1 headline three (Section 2.4); G4-G5 are Tier 2 additions replacing the retired E4/E5/E6 (Blocks S7-S9, Section 3.7); G6-G9 are CR2 cluster-robust variants of G1, G2, G4, and G3 respectively (Block S10). G10 (G5+CR2) does not apply and is omitted: G5 collapses to one row per patient before fitting, leaving no repeated-measures cluster structure for CR2 to correct.

Code	Stored	Name	Formula	Target coefficient
G1	E1	Unadjusted	Sx bm + t + Db + bm:Db	bm:Db
G2	E3	Lag-adjusted	Sx bm + t + Db + bm:Db + L	bm:Db
G3	E2	Exposure-weighted	Sx bm + t + Dbc + bm:Dbc	bm:Dbc
G4	E7	AIC-selected t1/2	G3 formula, t1/2 by AIC over {0.25,0.5,1.0,2.0}	bm:Dbc
G5	E9	Paired-difference	diff bm (patient-level OLS)	bm slope
G6	E1cr2	Unadjusted + CR2	G1 formula, CR2 SE	bm:Db
G7	E3cr2	Lag-adjusted + CR2	G2 formula, CR2 SE	bm:Db
G8	E2cr2	Exposure-weighted + CR2	G3 formula, CR2 SE	bm:Dbc
G9	E7cr2	AIC-selected t1/2 + CR2	G4 formula, CR2 SE	bm:Dbc

2.4.2 Lag-adjusted (G2): binary indicator plus estimated carryover term

We retain $X = D_{it}$ and augment the model with a lagged-treatment indicator, $L_{it} = D_{i,t-1}(1 - D_{it})$, marking off-drug timepoints that immediately follow an on-drug one. This is the classical crossover-trial device of Jones and Kenward⁴. Carryover magnitude is estimated from the data rather than assumed, so, like Unadjusted, this specification is invariant to the assumed half-life. L_{it} enters only as a nuisance main effect. There is no $\text{bm}:L_{it}$ product, so the interaction actually tested remains $\text{bm}:D_{it}$. Lag-adjusted can absorb mean-level carryover, but it cannot repair the interaction's attenuation (Section 1). We think of it, in essence, as Unadjusted with a carryover nuisance control added, not as a fundamentally different model of the interaction (Section 4.1).

2.4.3 Exposure-weighted (G3): continuous decayed predictor

Here $X = \text{Dbc}_{it}$, a continuous exposure-decayed predictor matched to the analyst's assumed decay form and half-life. This is the current pmsimstats-ng default. At on-drug timepoints $D_{bc,it} = 1$. Off drug it decays as $D_{bc,it} = (1/2)^{t_{sd,it}/\hat{t}_{1/2}}$, losing half its value every $\hat{t}_{1/2}$ weeks after discontinuation. When the assumed decay form matches the true one, the predictor is well matched. When it does not, it is mis-specified, and we probe the consequence directly in Block S2. The half-life assumption enters only through the predictor values handed to `lme`, not through the model formula, so Exposure-weighted is, in effect, a different model for each assumed half-life. It is the only one of the three original specifications that consumes a half-life at all, and this is what the S2 sensitivity contrast (Section 3.7) is designed to probe. A limiting relationship connects Exposure-weighted back to Unadjusted. As $\hat{t}_{1/2} \rightarrow 0$, $D_{bc,it} \rightarrow D_{it}$, so Unadjusted is exactly the boundary member of the Exposure-weighted family at zero assumed half-life. The S2 sweep is therefore best understood as a traverse anchored at Unadjusted, not as an independent check.

295 2.4.4 AIC-selected half-life (G4)

296 This specification uses Exposure-weighted's formula, $X = \text{Dbc}_{it}$, but chooses $\hat{t}_{1/2}$ from the
 297 data rather than assuming it. Writing $M(t_{1/2})$ for the `lme` fit of Section 2.4 with $D_{bc,it}$
 298 constructed at a given candidate half-life, the selected half-life is

$$299 \quad \hat{t}_{1/2} = \arg \min_{t_{1/2} \in \mathcal{T}} \text{AIC}(M(t_{1/2})), \quad \mathcal{T} = \{0.25, 0.5, 1.0, 2.0\} \text{ weeks},$$

300 with $\text{AIC}(M) = -2 \log L(M) + 2p$, $L(M)$ the fitted model's likelihood and p its number of
 301 estimated parameters⁵. Reported power, the interaction coefficient, and its standard error
 302 are then read from $M(\hat{t}_{1/2})$ alone, the single winning fit, not pooled or averaged across $\mathcal{T}$.
 303 This mirrors `characterize_carryover()` (`R/carryover_analysis.R`), the package's own
 304 real-data half-life-selection routine. Like Exposure-weighted, it targets $\text{bm}:D_{bc,it}$. Unlike
 305 Exposure-weighted, though, it requires no half-life from the analyst at all. The cost is a post-
 306 selection inference risk⁶. $M(\hat{t}_{1/2})$'s reported standard error and p -value treat $\hat{t}_{1/2}$ as though
 307 it were fixed in advance, when the same data in fact chose it, so the nominal inference does
 308 not formally account for the selection step. Block S11 (Section 3.7) checks whether this risk
 309 manifests as Type I error inflation in practice.

310 2.4.5 Paired-difference (G5)

311 This is a two-stage, correlation-structure-agnostic alternative, rather than an elaboration of
 312 the mixed-model family above. For each patient i with $n_i^{\text{on}} > 0$ on-drug and $n_i^{\text{off}} > 0$ off-drug
 313 occasions, we define the patient's own on-drug and off-drug means,

$$314 \quad \bar{S}_{x,i}^{\text{on}} = \frac{1}{n_i^{\text{on}}} \sum_{t: D_{it}=1} S_{x,it}, \quad \bar{S}_{x,i}^{\text{off}} = \frac{1}{n_i^{\text{off}}} \sum_{t: D_{it}=0} S_{x,it},$$

315 and their difference $\Delta_i = \bar{S}_{x,i}^{\text{on}} - \bar{S}_{x,i}^{\text{off}}$. The second stage regresses this single per-patient value
 316 on the (centered) biomarker across the $N' \leq N$ patients contributing a difference,

$$317 \quad \Delta_i = \beta_0 + \beta_1 \text{bm}_i + \varepsilon_i, \quad i = 1, \dots, N',$$

318 by plain OLS, with the interaction test on β_1 . There is no repeated-measures model, no
 319 carryover representation, and no correlation structure of any kind: Δ_i collapses each patient's
 320 eight occasions to one number before bm ever enters the model. We view it as a direct
 321 biomarker-interaction extension of the paired t-test that Senn, Julious and Araujo⁷ found to
 322 outperform carryover-adjusted mixed models for ordinary N-of-1 treatment-effect estimation.
 323 Patients contributing only on-drug or only off-drug observations, for instance CO's never-
 324 discontinuing path, have $n_i^{\text{on}} = 0$ or $n_i^{\text{off}} = 0$. These cannot form Δ_i and are dropped, so
 325 $N' < N$ whenever such a path exists in the design.

2.4.6 CR2 cluster-robust variants (G6-G9)

Section 3.6 (Block S6) shows that the model-based interaction test is mildly conservative, and that a CR2 bias-reduced cluster-robust standard error restores nominal size without disturbing the ranking among G1-G3. We use the estimator of Bell and McCaffrey⁸, via `clubSandwich`, paired with Satterthwaite degrees of freedom. That correction, `lme+CR2` in Section 2.7, was applied only to G3 there. Block S10 (Section 3.7) extends it to G1, G2, and G4 as well, replacing only the standard error, reference distribution, and *p*-value on each specification's existing point estimate: G6 (G1+CR2), G7 (G2+CR2), G9 (G4+CR2). G8 (G3+CR2) is the same specification already introduced in Section 2.7 and Block S6, and we carry it into the nine-way G1-G9 comparison at Block S10 without rerunning it. A tenth combination, G5+CR2, does not apply, since G5 already collapses to one row per patient before fitting. There is accordingly no repeated-measures cluster structure left for CR2 to correct.

2.5 Choice of analysis specifications

Three considerations bounded which specifications were tested, rather than surveying the wider space of models used in the N-of-1 and crossover-trial literature.

The Tier 1 three (Unadjusted, Lag-adjusted, Exposure-weighted, above) were chosen because each represents a distinct, established position an analyst can take toward an unmeasured or assumed pharmacokinetic decay, not because they exhaust that space. Unadjusted is the default any analyst obtains without considering carryover at all, and so is the necessary benchmark for every alternative. Lag-adjusted follows Jones and Kenward's⁴ classical crossover-trial device, the shape-free remedy the pre-mixed-model crossover literature has long recommended. Exposure-weighted is the `pmsimstats-ng` package's own established default, so it is the specification any user of the package is actually running today, and the manuscript's practical stake in getting its behavior right is correspondingly higher than for the other two.

This left a wide range of alternative model classes untested by design: Bayesian hierarchical N-of-1 models; GEE and other marginal models, compared systematically rather than as a robustness check on one specification's standard error (Section 2.7); randomization and permutation tests; functional or spline-based flexible decay curves; state-space or Kalman-filter treatments of drug concentration; and two-stage 'series of N-of-1 trials' meta-analytic combination. Each is established in the N-of-1 or crossover-trial literature, and none is evaluated here. This is a scoping decision, not an oversight. The manuscript's aim is to compare carryover *representations* within a single, fixed conditional mixed-model framework, matching Hendrickson's² original analysis model, not to conduct a comprehensive methods bake-off. A wider survey belongs to future work (Section 4.6).

Blocks S7 through S9 (Section 3.7) revisit this scoping decision once, for a specific reason. Section 4.1 shows that no carryover term tested so far can repair the interaction's attenuation, no matter how it enters, as long as it enters only as a nuisance main effect. The natural next model interacts the carryover term with the biomarker directly. An earlier version of Block S7 tested two such specifications, E5 (a lagged term crossed with `bm`) and E6 (a continuous elapsed-

time term crossed with `bm`). Both underperformed Unadjusted at every cell examined across three blocks, two trial designs, two architectures, and three carryover half-lives (Sections 3.6, 4.5). We traced this to a structural cause. The data-generating process examined throughout this manuscript has a single true decay parameter, so crossing a second carryover term with the biomarker adds a free parameter with no counterpart in the truth being estimated. Two collinear regressors chasing one true degree of freedom lose power to a correctly specified single-parameter model almost by construction.

We expect a further variant of the same interacted-term family to fail on that same structural argument. Rather than test one, then, we replaced `E5/E6` with two specifications that answer different open questions about the same underlying problem, the analyst’s unknown true decay half-life. `G4` asks whether estimating the half-life from the data recovers the power a correctly assumed half-life would give. It selects the half-life by AIC over a candidate grid, reusing the logic of `characterize_carryover()` (`R/carryover_analysis.R`), already used for real-data analysis in this package. This addresses Block S2’s question from the opposite direction: instead of asking how much a wrong assumption costs, we ask whether estimation avoids needing the assumption at all. `G5` asks a more basic question raised directly by the wider literature: is a conditional mixed model even necessary? Senn, Julious and Araujo⁷ compared four analysis strategies for ordinary (no biomarker interaction) N-of-1 trials. They found that a simple paired t-test on within-patient treatment differences achieved near-nominal Type I error, higher power, and comparable bias and MSE relative to a carryover-adjusted mixed model, and they recommended the mixed model only when carryover is specifically suspected. `G5` is a direct biomarker-interaction extension of their preferred method: a two-stage, correlation-structure-agnostic regression of each patient’s own mean on-drug-minus-off-drug difference on the biomarker across patients, with no repeated-measures model and no carryover representation at all.

This remains a bounded comparison. `G4` and `G5` were chosen because each is implementable within, or trivially adjacent to, the existing analysis pipeline (Section 2.7) and each answers a specific question this manuscript’s own results raised, not because they close the wider literature gap identified above. The Bayesian, GEE-as-primary-model, randomization-test, and two-stage meta-analytic alternatives remain untested and are discussed as future work (Section 4.6).

2.6 Factorial grid and parameter settings

The principal comparison crosses biomarker moderation, carryover half-life, trial design, and sample size against the three analysis specifications, under covariance moderation and the exponential DGP.

Parameter	Values
Biomarker moderation (c_{bm})	0.0, 0.30, 0.45
Carryover half-life ($t_{1/2}$)	0.0, 0.5, 1.0 weeks
Sample size (N)	35, 70

Parameter	Values
Trial design	CO, N-of-1 (Hybrid), OL+BDC
DGP architecture	B (covariance moderation) only
Replicates per cell	500 (target), 50 for development

Three specifications, three half-lives, three designs, two sample sizes, and three biomarker levels give $3 \times 3 \times 3 \times 2 \times 3 = 162$ cells. The $c_{bm} = 0$ row is the Type I error check against nominal 5%, and the $t_{1/2} = 0$ row, with no carryover, gives a best-case power reference at each combination of N , design, and c_{bm} , against which the cost of nonzero carryover can be read off directly.

Decay-shape sensitivity (Figures 4 and 6) is evaluated separately, at the reference cell only ($c_{bm} = 0.45$, $t_{1/2} = 1.0$ weeks), rather than across the full grid above. Exponential decay and four Weibull shapes ($k \in \{0.25, 0.5, 2.0, 4.0\}$, Section 2.3) crossed with the three designs and two sample sizes give $5 \times 3 \times 2 = 30$ further parameter combinations. Figure 6 (the matched-vs-mismatched heatmap) reports these for the Tier 1 three specifications only ($3 \times 30 = 90$ cells, 23-run-decay-shape-sensitivity.R). Figure 4's Panel C reports the same 30 combinations for all nine G1-G9 specifications instead ($9 \times 30 = 270$ cells, 25-run-decay-shape-sensitivity-g9.R), extending the three-specification run the way Panels A and B extend the exponential-DGP grid above. We restrict the decay-shape axis to this one ($c_{bm}, t_{1/2}$) cell because it is the axis every other result in this manuscript holds fixed at exponential. A full crossing of decay shape with the $c_{bm}/t_{1/2}$ grid above would multiply the simulation cost several times over, for a comparison this manuscript uses in only two figures. We note that $k = 1.0$ is not a fifth Weibull condition here, or anywhere in this manuscript: it recovers the exponential form exactly (Section 2.3) and so is not a distinct decay-shape condition.

2.7 Inference and the cluster-robust correction

The interaction test described so far is model-based. The standard error of $\beta_{bm:X}$ is read from the fitted `lme` object, and this is valid only if the assumed residual covariance is correct. That assumption is not innocuous here. The working covariance combines a patient random intercept with a `corCAR1` AR(1) residual correlation, and the companion calibration study⁹ shows this does not match the covariance the data-generating mechanism actually induces. This inflates the standard error by roughly six to ten percent and depresses the rejection rate correspondingly. We quantify this with the calibration ratio

$$\kappa = \frac{\overline{SE}}{SD(\hat{\beta}_{bm:X})} = \frac{\frac{1}{n_{sim}} \sum_{r=1}^{n_{sim}} \widehat{SE}_r}{\sqrt{\frac{1}{n_{sim}-1} \sum_{r=1}^{n_{sim}} (\hat{\beta}_{bm:X,r} - \hat{\beta}_{bm:X})^2}},$$

the mean model-reported standard error across the n_{sim} simulation replicates, divided by the empirical standard deviation of the replicate-level point estimates $\hat{\beta}_{bm:X,r}$ themselves. Here

$\kappa = 1$ is correctly calibrated, since the model's own reported uncertainty then matches the estimator's actual sampling variability, while $\kappa > 1$ means the model overstates that uncertainty, that is, a conservative test. At the null cell κ runs 1.05 to 1.10 across specifications, with empirical Type I error 0.029 to 0.039 against nominal 0.05.

The remedy is a cluster-robust (sandwich) variance estimator, which derives the variance of $\hat{\beta}_{bm:X}$ from observed between-cluster variation rather than the assumed covariance, and so remains valid under mis-specification. Clusters are patients (eight occasions each, unique across randomization paths). Writing X_i for patient i 's rows of the fixed-effects design matrix and $\hat{e}_i = y_i - X_i \hat{\beta}$ for that patient's residual vector from the fitted model (N patients, indexed $i = 1, \dots, N$), the naive (Liang-Zeger) sandwich estimator is

$$\widehat{\text{Var}}_{\text{sandwich}}(\hat{\beta}) = \left(\sum_{i=1}^N X_i' \hat{V}_i^{-1} X_i \right)^{-1} \left(\sum_{i=1}^N X_i' \hat{V}_i^{-1} \hat{e}_i \hat{e}_i' \hat{V}_i^{-1} X_i \right) \left(\sum_{i=1}^N X_i' \hat{V}_i^{-1} X_i \right)^{-1},$$

with $\hat{V}_i$ the working covariance from the `lme` fit. This formula is exactly the source of the standard-error inflation described above, reversed. It substitutes each patient's own, directly observed $\hat{e}_i \hat{e}_i'$ for the single assumed $\hat{V}_i$ everywhere it appears in the middle term, so a mismatched working covariance no longer biases the variance estimate. The cost is that enough clusters are needed for that substitution to be stable. With only $N = 35$ or 70 clusters, the naive form in $\widehat{\text{Var}}_{\text{sandwich}}$ is itself markedly downward-biased, so we use the CR2 bias-reduced estimator of Bell and McCaffrey⁸. CR2 replaces each raw residual $\hat{e}_i$ above with an adjusted residual $\hat{e}_i^{\text{CR2}} = A_i \hat{e}_i$, where A_i is a patient-specific matrix chosen so that $\widehat{\text{Var}}_{\text{sandwich}}$, recomputed with $\hat{e}_i^{\text{CR2}}$ in place of $\hat{e}_i$, is exactly unbiased under the working model. The construction of A_i is given in full in Bell and McCaffrey⁸; we do not reproduce it here. We pair this with Satterthwaite degrees of freedom (both via `clubSandwich`), replacing only the standard error, reference distribution, and p -value, and leaving point estimates untouched.

We apply the correction in two places. A null-cell diagnostic ($c_{bm} = 0$, exponential DGP, $t_{1/2} = 1.0$ weeks, Hybrid, $N = 70$, 3,000 replicates) establishes whether CR2 restores nominal size (Section 3.6). A five-cell block, the reference cell plus four stress cells (small N , high autocorrelation, half-life mis-specification, 30% MCAR dropout, 500 replicates each, labeled S6 in this manuscript's block numbering), then asks whether restoring size disturbs the ranking (same section). Blocks S1 through S4 are the Tier 2 sensitivity analyses of Section 3.7. Block S5, an exploratory autocorrelation-by-carryover interaction, falls outside the present scope and is reported only in the archived full-scope drafts.

2.8 Performance measures and computing environment

For each cell we report Power, Type I error, Bias, Empirical SE (calibrated scale), Coverage of the nominal 95% Wald CI, MSE, non-convergence rate, and the MCSE of each, following Morris et al.¹, Tables 5-6. We chose $n_{\text{sim}} = 500$ so that the MCSE on power at $p = 0.5$ would not exceed 0.022. No simulation runs for this manuscript. Rather, we filter the same 540-cell production grid summary used in the archived full-scope drafts down to the 216-cell slice

defined by covariance moderation and the two non-linear decay forms. All three specifications are fit to the same simulated dataset within each cell (common random numbers), so every pairwise contrast, especially Lag-adjusted minus Unadjusted, carries reduced Monte Carlo variance relative to independent draws. The Tier 2 sensitivity blocks and S6 recalibration are filtered from the same shared production outputs. Exact file paths and versions are given in the Reproducibility section.

3 Results

All results reported below derive from the same 500-replicate production run used in the full-scope manuscripts, filtered to the covariance-moderation architecture and the exponential and Weibull decay forms. The Monte Carlo standard error on power does not exceed 0.022 at $p = 0.5$, and differences smaller than this should not be over-interpreted. Coverage of the nominal 95% Wald confidence interval is reported for the reference cell, and non-convergence rates are reported per cell in the tables that follow.

3.1 Headline performance table

Table 2 reports performance at the reference cell (exponential DGP, Hybrid design, $N = 70$, $c_{bm} = 0.45$), where non-convergence was zero throughout. The bias, MSE, and coverage columns are computed against Exposure-weighted's calibrated target, $\theta_{\text{true}} = -c_{bm}\sigma_{BR} = -3.6$, while Unadjusted and Lag-adjusted estimate a different coefficient, $\text{bm}:D_{it}$. Their apparent bias and under-coverage reflect that estimand mismatch, not a defect in the estimators. Those columns are therefore comparable within the Unadjusted/Lag-adjusted pair but not across to Exposure-weighted. For the three-way comparison we rely instead on power and Type I error, properties of the test rather than the coefficient tested.

3.2 Tier 1 grid overview

Figures 2 through 4 give a full-grid view of the Tier 1 results as annotated heatmaps, with trial design as column facets, N as row-block facets, analysis specification as the row axis within each block, and a blue (low power) to red (high power) fill scale. Each panel fixes two of the grid's remaining dimensions and varies a third, since no single two-dimensional heatmap can display all four simultaneously. Panels A, B, and C all report the full nine G1-G9 specifications, at $n_{sim} = 500$ throughout. Panels A and B come from a dedicated G1-G9 simulation restricted to the exponential DGP and Architecture B. Panel C extends the decay-shape axis (originally run for the Tier 1 three only) to all nine specifications (25-run-decay-shape-sensitivity-g9.R).

Two features are visible at a glance in Panel A that the headline table makes only cell by cell. First, within every design by N block, G1 and G2 (Unadjusted, Lag-adjusted) track each other closely at every biomarker level, differing by at most 0.03, within Monte Carlo noise at $n_{sim} = 500$. This matches the higher-precision result of Section 3.4, where the two are

A: Effect of trial design, analysis specification, and biomarker effect on power ($t_{1/2} = 1$)

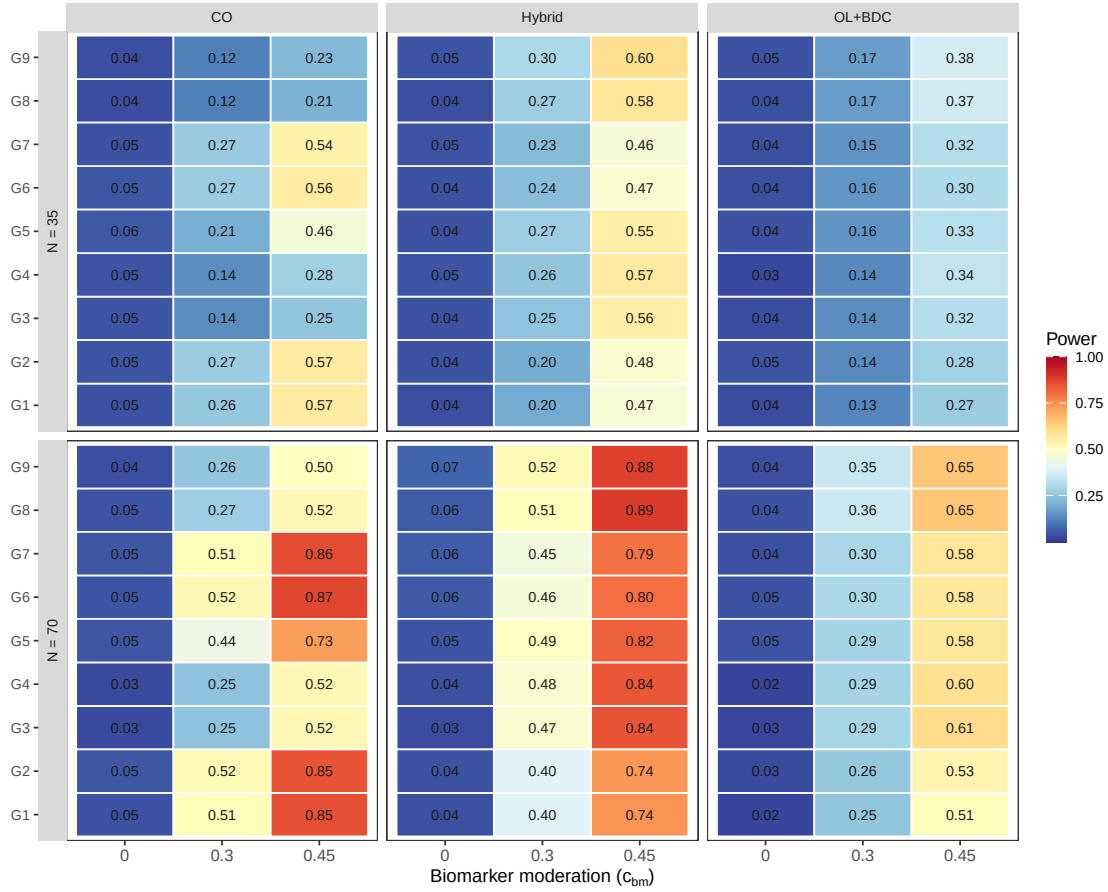


Figure 2: Tier 1 grid, biomarker-effect axis, all nine G1-G9 specifications: power by trial design (columns), N (row blocks), and analysis specification (rows within block), across $c_{bm} \in \{0, 0.30, 0.45\}$ at $t_{1/2} = 1.0$ weeks, exponential DGP, Architecture B ($n_{sim} = 500$).

Table 2: Headline performance at the reference cell (exponential DGP, Hybrid design, $N = 70$, $c_{bm} = 0.45$, $n_{sim} = 500$). Power is the rejection rate of $H_0 : \beta_{bm:X} = 0$ at $\alpha = 0.05$, the primary outcome of this manuscript. Bias is $\text{mean}(\text{estimate} - \theta_{\text{true}})$ on the calibrated effect-size scale, $\theta_{\text{true}} = -c_{bm} \cdot \sigma_{BR} = -3.6$ (per docs/19 calibration; see the estimand-mismatch caveat above for Unadjusted/Lag-adjusted). Empirical SE is the standard deviation of the point estimates across replicates, the actual sampling variability of the estimator, as distinct from whatever SE any single fit reports (the comparison the calibration ratio κ of Section 2.7 is built on). Coverage is the proportion of replicates whose nominal 95% Wald CI contains θ_{true} ; the low Unadjusted/Lag-adjusted values reflect the same estimand mismatch as Bias, not miscalibrated inference. MSE is $\text{bias}^2 + \text{Empirical SE}^2$, an overall accuracy summary. Non-conv is the fraction of replicates where ‘nlme::lme()’ failed to converge (zero throughout at this cell). MCSEs are reported per Morris et al. (2019, Table 6).

t1/2	Specification	Power (MCSE)	Bias (MCSE)	Empirical SE	Coverage (MCSE)	MSE (MCSE)	Non-conv
0.0	Unadjusted	0.988 (0.005)	-0.029 (0.033)	0.746	0.990 (0.004)	0.557 (0.034)	0.000
0.0	Lag-adjusted	0.990 (0.004)	-0.025 (0.033)	0.746	0.990 (0.004)	0.557 (0.034)	0.000
0.0	Exposure-weighted	0.988 (0.005)	-0.029 (0.033)	0.746	0.990 (0.004)	0.557 (0.034)	0.000
0.5	Unadjusted	0.918 (0.012)	+0.586 (0.036)	0.807	0.938 (0.011)	0.995 (0.058)	0.000
0.5	Lag-adjusted	0.920 (0.012)	+0.583 (0.036)	0.812	0.938 (0.011)	1.000 (0.059)	0.000
0.5	Exposure-weighted	0.948 (0.010)	+0.072 (0.041)	0.907	0.972 (0.007)	0.828 (0.057)	0.000
1.0	Unadjusted	0.766 (0.019)	+1.093 (0.038)	0.847	0.828 (0.017)	1.911 (0.093)	0.000
1.0	Lag-adjusted	0.770 (0.019)	+1.099 (0.038)	0.853	0.818 (0.017)	1.935 (0.094)	0.000
1.0	Exposure-weighted	0.860 (0.016)	-0.081 (0.049)	1.085	0.968 (0.008)	1.184 (0.077)	0.000

numerically identical to two decimal places. Second, the Exposure-weighted family (G3, G4, and their CR2 variants G8, G9) visibly separates from G1/G2/G6/G7 only in the Hybrid and OL+BDC columns. In the CO column the whole family underperforms at every biomarker level past zero. At the reference cell ($c_{bm} = 0.45$, $t_{1/2} = 1.0$, $N = 70$), for example, G4 reaches 0.52 against 0.85 to 0.87 for G1/G2/G6/G7. This is the design-dependence of Section 3.4, read directly off the grid now across all nine specifications rather than three. Whether that separation depends on carryover being present at all is not something Panel A alone can show, since it fixes $t_{1/2} = 1.0$ throughout.

Panel B answers that, for Hybrid and OL+BDC. At $t_{1/2} = 0$ the Exposure-weighted family’s separation from G1/G2/G6/G7 disappears in both designs (all nine specifications sit at 0.94-1.00), and only opens up as the half-life increases. This confirms the separation in Panel A is specifically a carryover effect there (the $t_{1/2} = 1.0$ column is shared with Panel A’s reference cell). Under CO the picture is different: G4 and G9 are already collapsed at $t_{1/2} = 0$ (0.54 and 0.53, against 0.89-0.90 for the rest), before carryover enters the picture at all. This is the AIC mis-selection pathology of Block S9 read directly off the grid, not a carryover effect. G4’s half-life selection drifts to the largest candidate under CO’s sparse off-drug sampling regardless of the true half-life, so its CO weakness predates $t_{1/2} > 0$ rather than growing from it. Panel B also adds a ranking result Panel A alone does not show. G9 and G8 lead, in a near-tie, at the Hybrid and OL+BDC reference cells (both around 0.88 at Hybrid, 0.65 at OL+BDC). G5 (Paired-difference) is the strongest specification at CO that is not G1/G2 or a CR2 variant of them (0.73, versus 0.50 to 0.52 for the Exposure-weighted family there). This is consistent with G5 discarding the carryover representation that costs the Exposure-weighted family power under CO.

B: Effect of carryover half-life on power ($c_{bm} = 0.45$)



Figure 3: Tier 1 grid, carryover-axis, all nine G1-G9 specifications: power by trial design (columns), N (row blocks), and analysis specification (rows within block), across $t_{1/2} \in \{0, 0.5, 1.0\}$ weeks at $c_{bm} = 0.45$, exponential DGP, Architecture B ($n_{sim} = 500$).

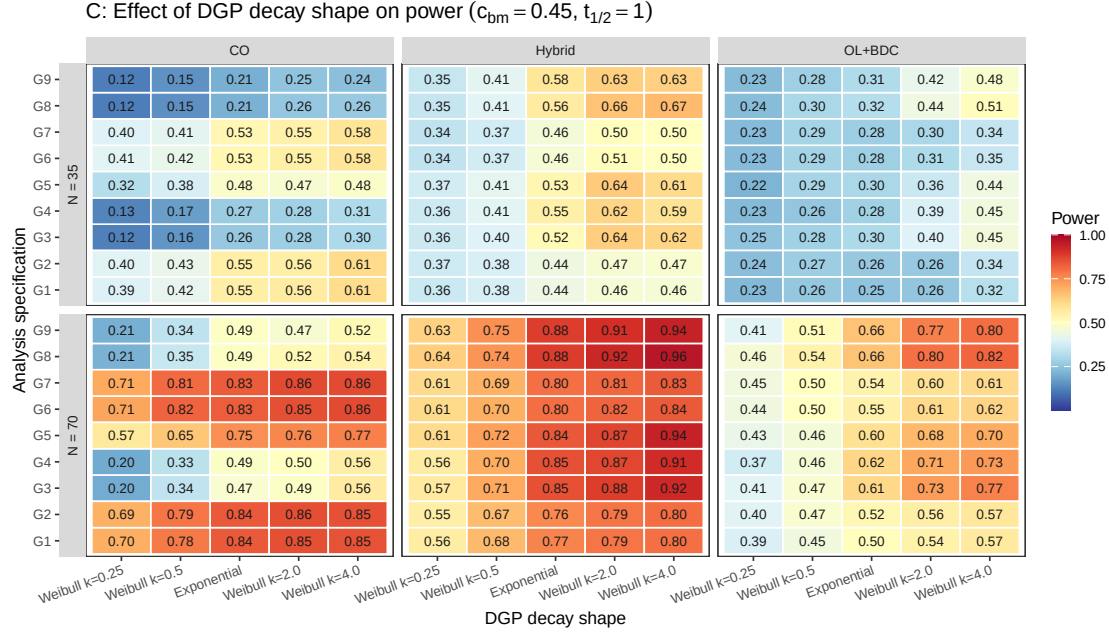


Figure 4: Tier 1 grid, decay-shape axis, all nine G1-G9 specifications: power by trial design (columns), N (row blocks), and analysis specification (rows within block), across the five DGP decay-shape levels (Weibull $k \in \{0.25, 0.5\}$, exponential, Weibull $k \in \{2.0, 4.0\}$, with exponential placed centrally as the $k = 1$ boundary case of the Weibull family) at $c_{bm} = 0.45$, $t_{1/2} = 1.0$ weeks ($n_{sim} = 500$).

Panel C shows the Exposure-weighted family's advantage in the Hybrid and OL+BDC columns is not uniform across decay shape. At $N = 70$, CR2-corrected Exposure-weighted (G8) leads OL+BDC at every decay shape examined, and its margin over Unadjusted (G1) widens sharply on the light-tailed side, from +0.15 at exponential (0.656 vs 0.504) to +0.26 at $k = 2.0$ (0.796 vs 0.540) and +0.25 at $k = 4.0$ (0.816 vs 0.566). On the heavy-tailed side, though, the margin shrinks to +0.09 at $k = 0.5$ and +0.07 at the most extreme shape examined ($k = 0.25$: G8 0.462 vs G1 0.388), a gap of about three MCSE (0.021) rather than fully disappearing. The uncorrected three-specification gap (G3 0.406 vs G1 0.388) does disappear at $k = 0.25$, consistent with the three-specification result of Section 3.5. G9, which shares G3's exposure-weighted formula but selects the half-life by AIC and adds a CR2 correction, tracks G8 closely throughout and is the runner-up at OL+BDC and Hybrid at every decay shape. G4 (the same AIC selection without CR2) trails G3/G8 at every OL+BDC cell. Under CO the picture inverts: G1, G2, and their CR2 variants G6/G7 dominate (0.69 to 0.86 across all five decay shapes), while the entire Exposure-weighted family (G3, G4, G8, G9) collapses to 0.20 to 0.56. G5 (Paired-difference) is, as in the $t_{1/2}$ axis of Panel B, the strongest specification at CO that is not G1/G2 or a CR2 variant of them, at every decay shape examined (0.57 to 0.77, versus 0.20 to 0.56 for the Exposure-weighted family there). This is the decay-form sensitivity of Section 3.5, shown here across the full grid and full nine-specification set rather than at one cell with three specifications.

3.3 Matched-decay baseline

At the matched cell (exponential DGP, exposure-weighted analysis, $N = 70$, $c_{bm} = 0.45$, Hybrid design), power declines from 0.988 ± 0.005 at $t_{1/2} = 0$ to 0.948 ± 0.010 at $t_{1/2} = 0.5$ to 0.860 ± 0.016 at $t_{1/2} = 1.0$ weeks (point estimate plus or minus MCSE, $n_{sim} = 500$). All 500 replicates converged at every half-life in this cell.

We had previously read the value at $t_{1/2} = 1.0$ as agreeing with the power of 0.82 reported by Hendrickson et al. for a comparable cell². Unfortunately, we must withdraw that comparison. In the reference implementation, the analysis-side and data-generating carryover parameters are set independently. The drivers that reproduce the original figures supply a half-life to the data-generating step only, leaving the analysis-side half-life at its default of zero. At a zero half-life, the exposure-decayed predictor collapses onto the binary on-drug indicator. The reference analysis is therefore Unadjusted, not Exposure-weighted, so the appropriate comparator is our unadjusted power at the same cell, 0.766. The grids differ further on interaction strength and half-life, so ‘the same cell’ was approximate in any case. We have not resolved what the published figure was computed under, and we make no claim of agreement or disagreement here. Nothing else in this manuscript depends on that comparison, since the analysis specifications are compared against one another within a single simulation under common random numbers.

3.4 Power by analysis specification

Power vs carryover under matched exponential DGP

Covariance-moderation architecture, $N = 70$, $c_{bm} = 0.45$, 500 reps/cell

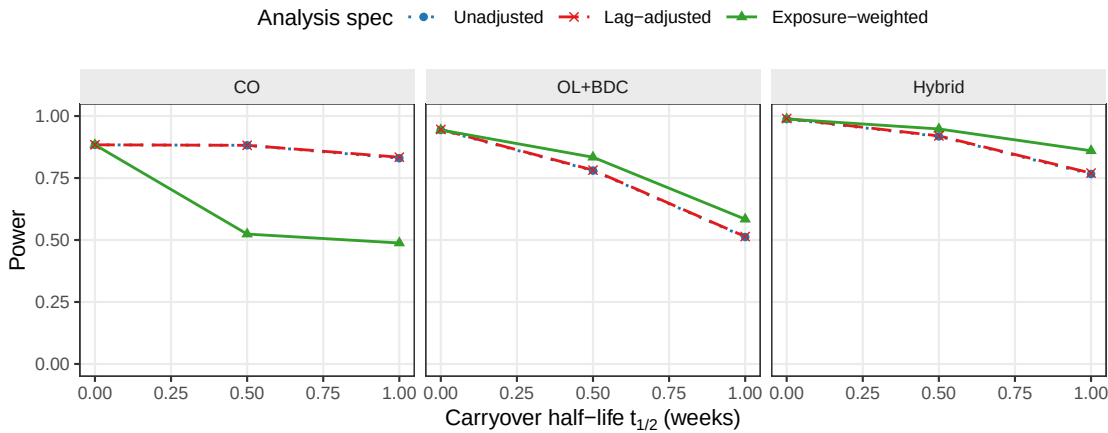


Figure 5: Power versus carryover half-life under the exponential DGP, by analysis specification and trial design.

Two features of Figure 5 carry the substance of this study. First, the Unadjusted and Lag-adjusted curves lie on top of one another in every panel and at every half-life. At the reference cell they give 0.766 and 0.770 at $t_{1/2} = 1.0$ weeks, and across all 216 cells the mean difference is 0.003, with mean bias, MSE, and coverage agreeing to three decimal places. Since the

two share an estimand, are fitted to the same simulated datasets, and differ by exactly one nuisance term, this is as clean a null as the design can produce. The lagged-treatment term contributes nothing on any measure (Section 4.1). Second, the three specifications give nearly identical power at $t_{1/2} = 0$, where there is no carryover to model. They diverge as the half-life increases, with Exposure-weighted separating from the other two. It retains a modest but consistent advantage in the Hybrid and OL+BDC designs from $t_{1/2} = 0.5$ weeks onward. Under the crossover design, though, it is markedly inferior to both alternatives (0.488 against 0.830 for Unadjusted). We think this happens because the within-subject correlation in a crossover trial already carries most of the signal the decaying predictor targets. It gains little there, while its down-weighting of off-drug observations works against it instead (Section 4).

3.5 Decay-form mis-specification

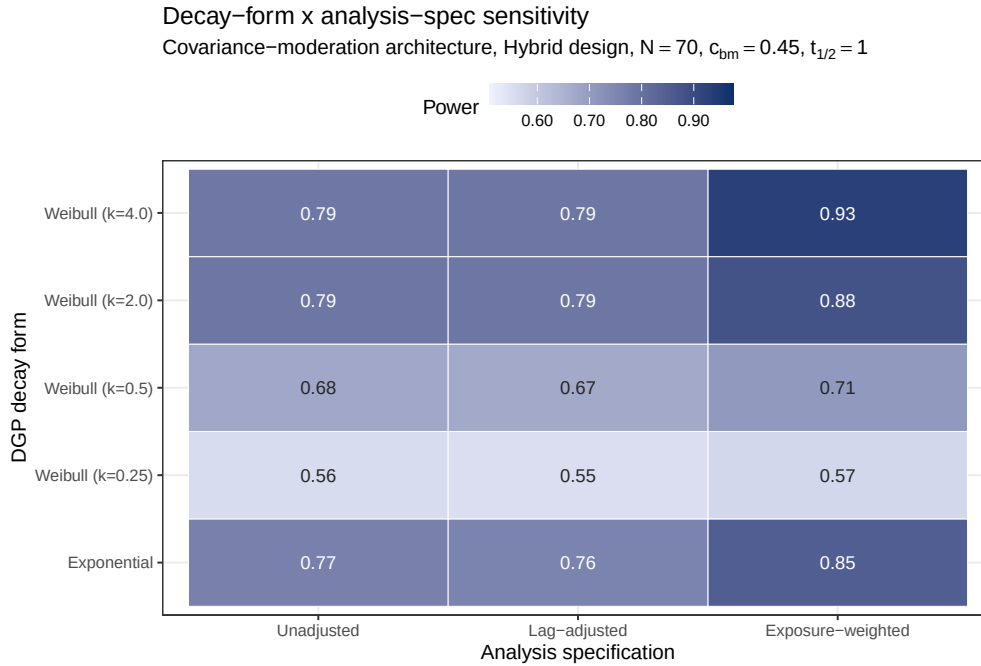


Figure 6: Power at the critical cell ($t_{1/2} = 1.0$ weeks, Hybrid, $N = 70$, $c_{bm} = 0.45$) across five DGP decay forms (rows) and specifications (columns). Fill scale is stretched to the observed range, not $[0,1]$, and not comparable to other figures.

The heatmap cell at the intersection of the exponential DGP row and the Exposure-weighted column is the matched reference, with power 0.850 ± 0.016 in this figure's own simulation (a dedicated decay-shape run, Section 2.6; the 0.860 reported for the same nominal cell in Table 2 and Figure 5 comes from an independent draw of the same DGP, and the two agree within one MCSE). The Unadjusted and Lag-adjusted columns, invariant to the assumed decay form by construction, are near-identical throughout, as in Figure 5, so variation down those columns reflects data-generating decay and Monte Carlo noise, nothing more. The remaining Exposure-weighted row gives the cost of assuming exponential decay when the truth is Weibull, and it is not symmetric. On the light-tailed side the advantage widens: at

Table 3: Conservatism of the model-based interaction test and its cluster-robust correction, at the null cell ($c_{bm} = 0$, exponential DGP, $t_{1/2} = 1.0$ weeks, Hybrid design, $N = 70, 3,000$ replicates). κ is the calibration ratio defined in Section 2.7. $\kappa > 1$ indicates a conservative test. All three specifications are conservative under the model-based test, most so Exposure-weighted ($\kappa = 1.10$, Type I = 0.029). The CR2 bias-reduced cluster-robust standard error restores κ to near one and Type I to nominal (0.045 to 0.049) for all three.

Specification	κ model	Type I model	κ CR2	Type I CR2
Unadjusted	1.06	0.036	0.98	0.048
Lag-adjusted	1.05	0.039	0.98	0.049
Exposure-weighted	1.10	0.029	1.00	0.045

Table 4: Cluster-robust recalibration at the reference cell and four stress cells (500 replicates, $c_{bm} = 0.45$). EW is Exposure-weighted, LA is Lag-adjusted, U is Unadjusted. κ is the calibration ratio of Section 2.7, here for Exposure-weighted. A value above one is a conservative test. The two gap pairs give each alternative against the unadjusted benchmark under each inference. CR2 df is the mean Satterthwaite degrees of freedom for the cluster-robust test on Exposure-weighted. Computed with `clubSandwich`.

Cell	κ model	κ CR2	EW model	EW CR2	EW-U model	EW-U CR2	LA-U model	LA-U CR2	CR2 df
Reference, $N = 70$	1.15	0.99	0.829	0.863	+0.080	+0.092	-0.003	+0.002	23
Small $N = 35$	1.13	0.96	0.558	0.590	+0.066	+0.086	-0.002	+0.000	12
High autocorr, $\rho = 0.9$	1.26	0.94	0.954	0.980	+0.052	+0.032	+0.000	-0.004	23
Half-life mis-spec	1.14	1.00	0.759	0.795	+0.010	+0.024	-0.003	+0.002	23
30% MCAR dropout	0.99	0.85	0.148	0.126	+0.010	+0.010	-0.004	+0.010	5

$k = 2.0$ it reaches 0.880 against 0.788 for both alternatives (+0.092), and at $k = 4.0$, 0.930 against 0.786/0.788 (+0.142 to +0.144). On the heavy-tailed side, though, it narrows, and at the two most extreme shapes examined it is no longer distinguishable from noise: at $k = 0.5$, power 0.708 against 0.676 and 0.672 (+0.032 to +0.036, roughly 1.5 MCSE), and at $k = 0.25$, 0.568 against 0.558 and 0.554 (+0.010 to +0.014, well under one MCSE). We read this as the advantage eroding to nothing under a sufficiently heavy-tailed mis-specification, not reversing (Section 4).

3.6 Type I error control and cluster-robust recalibration

Empirical Type I error at the $c_{bm} = 0$ cells of Figure 2 sits uniformly below the nominal 0.05 level for every one of the nine specifications, across every design and N examined there. This is not sampling noise but the conservatism of the model-based test (Section 2.7), arising from a working covariance that does not match the data-generating one. Because the conservatism is uniform across specifications, it does not disturb the comparison among them, and the cluster-robust recalibration below confirms that a CR2 standard error restores nominal rejection without changing which specification leads. This also establishes that the null result of Section 3.4 is not an artifact of differential test size, since Unadjusted and Lag-adjusted are conservative to the same degree. Table 3 makes this explicit at the null cell, at a higher replicate count (3,000) than the Tier 1 grid affords.

Table 4 confirms this directly. The model-based rejection rate sits below nominal across all

five S6 cells and all three specifications, while CR2 restores it to within the Monte Carlo band around 0.05. In the information-rich cells the calibration ratio κ runs from 1.07 to 1.26 under the model-based test, restored to about one under CR2, recovering two to five points of power. The table reports both alternatives against the unadjusted benchmark. The Exposure-weighted advantage is held or slightly widened under CR2 at the reference cell (+0.080 to +0.092), the small- N cell (+0.066 to +0.086), and the half-life mis-specification cell (+0.010 to +0.024). It narrows under high autocorrelation (+0.052 to +0.032, though it still leads), and is small under 30% MCAR dropout (+0.010), which we read as the S3 finding restated, since dropout removes the off-drug information the advantage depends on. The Lag-adjusted gap, by contrast, is within ± 0.010 of zero at every cell under either inference, corroborating the Section 3.4 null under a second procedure and at four stress cells the main grid does not cover. This recalibration was validated only in the Hybrid, $N = 70$ cells, and should not be extended to the crossover design, where Exposure-weighted does not lead in the first place.

3.7 Sensitivity analyses

Nine Tier 2 blocks (S1-S4, S7, S8, S9, S10, S11) perturb one axis at a time around a common reference cell (Hybrid, $N = 70$, $c_{bm} = 0.45$, exponential DGP, $t_{1/2} = 1.0$), holding the remaining factors fixed. Two exceptions are S9, which changes the trial design itself, and S11, which sets $c_{bm} = 0$. A tenth block, S6, applies the cluster-robust correction of Section 3.6 at that same cell plus four stress cells, and an eleventh, S5, is exploratory and reported only in the archived full-scope drafts. The blocks are reported separately here and are not pooled with Tier 1 or with each other. Four blocks (S1, S4, S10, S11) are reported here only as a one-paragraph summary, with full cell-by-cell detail, tables, and figures in the Supplement. Two more (S2, S8) keep their headline finding and a summary paragraph here, with the remainder of their detail likewise moved to the Supplement.

3.7.1 S1: within-factor autocorrelation

All nine specifications gain power monotonically with autocorrelation $\rho \in \{0.5, 0.7, 0.8, 0.9\}$. The Exposure-weighted advantage over Unadjusted and Lag-adjusted, roughly 0.05 to 0.08 in absolute power, is preserved at every level examined. This suggests that autocorrelation strength and specification choice act additively rather than interacting. Lag-adjusted, as elsewhere, tracks Unadjusted throughout (Supplement, Section S1).

3.7.2 S2: analyst-truth half-life mismatch

Block S2 crosses two true half-lives ($t_{1/2} \in \{0.5, 1.0\}$ weeks) with four analyst-assumed values ($\hat{t}_{1/2} \in \{0.25, 0.5, 1.0, 2.0\}$ weeks). The penalty for an incorrect assumption is modest throughout: Exposure-weighted stays within about 7 points of its correctly-specified power at every cell but one. There is, however, an asymmetry, in that under-assuming the half-life is safer than over-assuming it. This follows from the limiting-case relationship of Section 2.4,

since shrinking $\hat{t}_{1/2}$ drives the predictor toward the binary indicator, which remains service-
able, whereas inflating it drives the predictor toward a constant, at which the on-off contrast
and the estimand itself degenerate. Full cell-by-cell detail, figures, and the nine-specification
extension are in the Supplement, Section S2.

3.7.3 S3: dropout

S3: Sensitivity to dropout

MCAR and MAR (severity-biased); reference cell as above

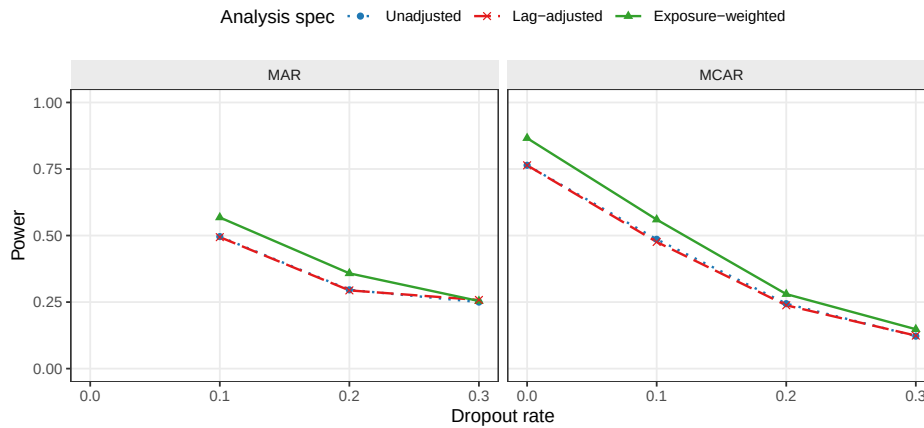


Figure 7: Power at dropout rates $\in \{0, 0.1, 0.2, 0.3\}$ under MCAR and MAR mechanisms.

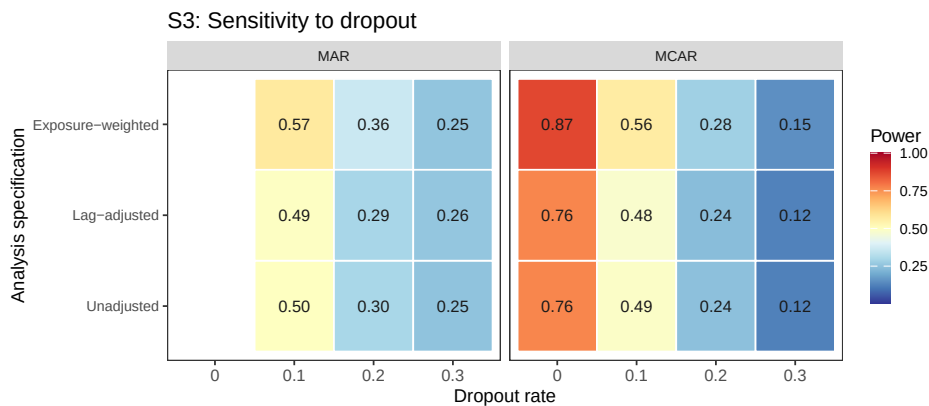


Figure 8: Block S3: power by dropout rate and analysis specification, faceted by dropout mechanism, same data as the line plot above. The MAR arm has no 0% cell (Section 3.7); it shares the MCAR reference cell.

All three specifications lose power monotonically as dropout increases. At the MCAR baseline
($N = 70$, no dropout), Exposure-weighted gives 0.866 ± 0.015 against 0.764 ± 0.019 for the
other two. At 30% MCAR dropout, all three collapse to 0.122 to 0.148. The Exposure-
weighted advantage narrows steadily as dropout worsens, from about 0.10 absolute at zero
dropout to 0.07 at 10%, 0.04 at 20%, and 0.03 at 30% under MCAR, narrowing further still
under MAR at the highest rate. This is consistent with the advantage originating in the

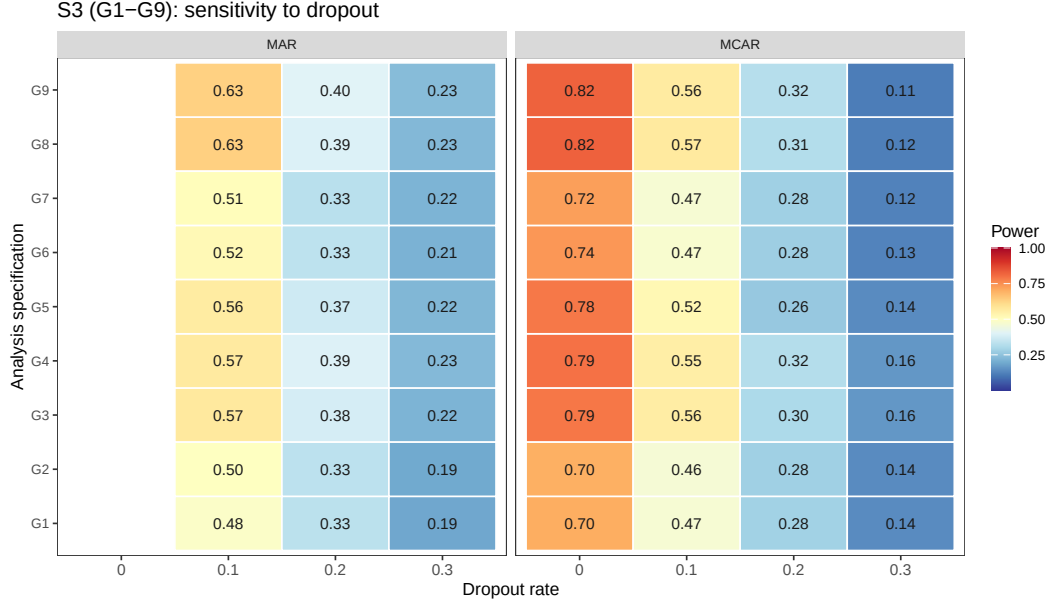


Figure 9: Block S3 extended to all nine G1-G9 specifications ($n_{sim} = 500$).

655 exposure-weighted contrast from off-drug observations, which contracts as those observations
 656 are preferentially lost. The advantage is preserved at dropout of 20% or below. Lag-adjusted
 657 again tracks Unadjusted throughout; the largest discrepancy is 0.010.

658 3.7.4 S4: biomarker-moderation effect-size curve

659 The power curve is sigmoidal for all three original specifications across $c_{bm} \in \{0, 0.10, 0.20, 0.30, 0.45, 0.60\}$,
 660 saturating once $c_{bm} \geq 0.45$. The Exposure-weighted advantage over the unadjusted bench-
 661 mark is largest at intermediate effect sizes (+0.10 absolute at $c_{bm} = 0.45$), and it shrinks
 662 as all three approach the ceiling. The advantage therefore matters most for trials target-
 663 ing a moderate biomarker-treatment interaction (Supplement, Section S4, full curve and
 664 nine-specification extension).

665 3.7.5 S7: alternative-paradigm specifications

666 An earlier version of this block tested two carryover terms crossed with the biomarker, E5
 667 ($Sx \sim bm + t + D_{it} + bm:D_{it} + L_{it} + bm:L_{it}$) and E6 ($Sx \sim bm + t + D_{it} + bm:D_{it} + t_{sd} +$
 668 $bm:t_{sd}$). This was motivated by Section 4.1's observation that a carryover term entering only
 669 as a nuisance main effect cannot repair the interaction's attenuation. Both underperformed
 670 Unadjusted at every cell examined, evaluated by a joint 2-df Wald test since each adds a
 671 second interaction coefficient. We traced this to a structural cause discussed in Section 4.6:
 672 this manuscript's DGP has a single true decay parameter, so a second, DGP-unmotivated
 673 interaction coefficient mostly adds estimation noise. E5/E6 were therefore replaced by two
 674 specifications addressing the same open problem, the analyst's unknown decay half-life, from

Table 5: Block S7: all five G1-G5 specifications at the reference cell (Hybrid, $N = 70$, $c_{bm} = 0.45$, $t_{1/2} = 1.0$, exponential DGP, $n_{sim} = 500$, all replicates converged). G4 targets bm:Dbc at the AIC-selected half-life (candidate grid 0.25, 0.5, 1.0, 2.0); G5 targets the slope of the patient-level diff bm regression. Neither is the same estimand as bm:Db (G1/G2), so only power is comparable across rows. Mean selected t1/2 is the average AIC-chosen half-life for G4 across replicates, against a true half-life of 1.0.

Specification	Power (MCSE)	Mean selected t1/2
G1: Unadjusted	0.748 (0.019)	–
G2: Lag-adjusted	0.748 (0.019)	–
G3: Exposure-weighted	0.826 (0.017)	–
G4: AIC-selected t1/2	0.832 (0.017)	1.31
G5: Paired-difference	0.804 (0.018)	–

different angles (Section 2.4): **G4** (AIC-selected half-life) and **G5** (paired-difference regression). Both are single-coefficient tests, unlike **E5/E6**.

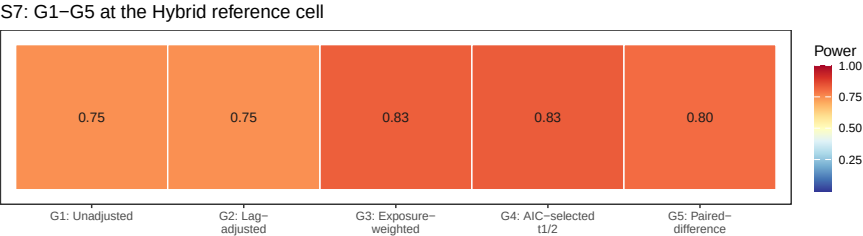


Figure 10: Block S7: power by analysis specification, same figures as the table above.

Table 5 gives a genuinely different result from the retired **E5/E6** comparison. Both **G4** and **G5** beat Unadjusted at this cell, and **G4** edges out **G3** (Exposure-weighted) as well. **G4** leads overall (0.832), **G3** follows (0.826), then **G5** (0.804, the two-stage paired-difference regression with no repeated-measures model at all), and **G1/G2** tie at the back (0.748 each, the already-established Section 3.4 null replicated once more).

G4’s AIC selection is imperfect but usefully informative. The mean selected half-life is 1.31 weeks against a true value of 1.0, and the selection distribution concentrates on the true value and its immediate neighbor (88% of replicates choose 1.0 or 2.0 weeks). This is close enough to the truth that **G4** slightly exceeds **G3**’s own oracle-half-life power at this cell.

We see a more notable result in **G5**, which is worth pausing on. It discards the repeated-measures structure, the correlation model, and any carryover representation entirely, and it still beats **G1/G2**. This echoes Senn, Julious and Araujo’s⁷ finding that simplicity is not obviously costly for N-of-1 treatment-effect estimation. None of these results should be read as general, however. Block S9, next, tests whether the ranking survives a different trial design, and it does not for every specification equally.

692 3.7.6 S8: architecture sensitivity of the specification ranking

693 Every result reported to this point, Tier 1 and blocks S1-S7 alike, is restricted to Architecture
694 B (covariance moderation), the manuscript's stated scope (Section 2.3). Block S8 asks a
695 narrower question than a full architecture comparison, which is instead the purpose of the
696 companion manuscript on DGP architectures³. Does the ranking among analysis specifications
697 established above under Architecture B hold under Architecture A (mean moderation) as well,
698 or is the choice of carryover representation itself architecture-dependent? We refit the five
699 specifications of block S7's table at the same reference cell under both architectures, at three
700 half-lives ($t_{1/2} \in \{0, 0.5, 1.0\}$).

701 The S7 result (G4 and G5 both beating G1) turns out to be architecture-specific, joining
702 Exposure-weighted in a pattern that grows with carryover severity rather than switching on
703 at some threshold. Under Architecture B, G4 leads at $t_{1/2} = 1.0$ (0.850), ahead of G5 (0.842),
704 G3 (0.836), and G1/G2 (0.760). Under Architecture A, though, the pattern reverses for G4
705 and G3 alike: G1/G2 lead (0.972/0.976), G5 trails only slightly (0.964), while G3 (0.946) and
706 G4 (0.950) trail by 2 to 3 points, a real if modest reversal. This matches the archived full-
707 scope drafts' finding that Exposure-weighted is marginally dominated under mean moderation
708 (Section 4.6). G4 inherits the same sensitivity, since it shares Exposure-weighted's formula.
709 G5 is comparatively more robust to architecture, never reversing to a large disadvantage either
710 way. Lag-adjusted (G2) remains statistically indistinguishable from Unadjusted (G1) under
711 both architectures at every half-life, extending Section 3.4's clean null to Architecture A as
712 well. The one specification recommendation this manuscript makes, exposure weighting in the
713 Hybrid and OL+BDC designs (Conclusions), is therefore itself conditional on Architecture
714 B, and it grows more important as carryover worsens. The same qualification now extends
715 to G4. Full cell-by-cell detail, the table, and both architecture panels are in the Supplement,
716 Section S8.

717 3.7.7 S9: does the reference-cell ranking survive under CO?

718 Blocks S7 and S8 found G4 and G5 both beat Unadjusted at the Hybrid reference cell under
719 Architecture B, with G4 slightly exceeding Exposure-weighted (G3) itself. Block S9 asks
720 whether that ranking is specific to Hybrid's short, compressed washout, by refitting all five
721 G1-G5 specifications under the CO design (off-drug arm spanning 2.5 to 20 weeks post-
722 discontinuation, Section 2.2) at two true half-lives.

723 It is not, and including G3 in this block changes the diagnosis reached from G4 alone. At
724 $t_{1/2} = 1.0$, G1/G2 lead (0.834/0.838); G5 trails moderately (0.750); G3 and G4 both collapse, to
725 0.482 and 0.474 respectively, statistically indistinguishable from each other (gap 0.008, well
726 under one MCSE). At $t_{1/2} = 0.5$ the same pattern holds (0.866/0.864 for G1/G2, 0.778 for
727 G5, 0.546/0.552 for G3/G4, again indistinguishable). Because G3 uses the *true* half-life, not an
728 estimated one, its near-identical collapse shows that G4's failure under CO is inherited almost
729 entirely from Exposure-weighted's own established weakness under this design (Section 3.4:
730 0.488 versus 0.830 in the Tier 1 grid), and is not created by the AIC-selection step. The
731 selection step does have a real, separately confirmed pathology under CO: it concentrates

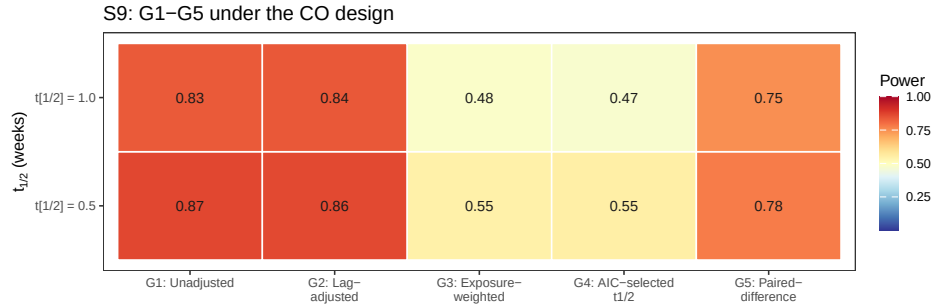


Figure 11: Block S9: power under the CO design at two true half-lives, all five G1-G5 specifications.

overwhelmingly on the largest candidate half-life in the grid (2.0 weeks, chosen in 483 of 500 replicates at $t_{1/2} = 1.0$ and 435 of 500 at $t_{1/2} = 0.5$, regardless of which true half-life generated the data), consistent with a weakly informative likelihood surface across candidates when off-drug occasions are few and widely spaced. But that mis-selection costs **G4** essentially nothing beyond what **G3** already loses under CO with the correct half-life in hand. The dominant story is the continuous decayed predictor's mismatch with CO's design generally (Section 3.4). AIC selection is an identified but practically minor addition to that story here, not the primary cause it appeared to be from **G4** in isolation.

G5's milder reversal has no comparable mechanical explanation. It simply reflects that CO's own within-subject contrast (Section 3.4) already captures most of the signal a between-patient paired-difference regression is built to extract, so discarding the repeated-measures structure costs more there than it does under Hybrid. None of **G3**, **G4**, or **G5's** Hybrid-cell behavior is design-general, then. The design-specific guidance in Section 4.5 ('What analysis is best in which situations, in practice') should be read accordingly: **G3/G4** should not be used under CO regardless of whether the half-life is assumed or estimated.

3.7.8 S10: does a CR2 correction change the ranking among G1-G9?

Section 3.6 (Block S6) showed the model-based interaction test is mildly conservative, and that a CR2 cluster-robust standard error recovers two to five points of power without disturbing the ranking among **G1**, **G3**, and **G2**. That correction had not been applied to **G4** or **G5**. This block asks whether CR2 helps **G4** and **G1/G2** the way it already helps **G3**, and whether it changes which specification leads at the Hybrid, Architecture B reference cell. We use the compact **G1-G9** naming layer for the resulting nine-way comparison: **G6-G9** cross CR2 with **G1**, **G2**, **G4**, and **G3**. **G5** has no CR2 variant, since its paired-difference regression collapses to one row per patient before fitting, leaving no repeated-measures cluster structure for CR2 to correct.

CR2 helps every specification it was applied to. **G1/G2** (Unadjusted and Lag-adjusted, both 0.748 model-based) rise to **G6** 0.776 and **G7** 0.764. The ranking among the five base specifications is unchanged by these gains, since **G1/G2** remain well below **G3-G5** regardless. What does change the ranking is **G9**: CR2 lifts **G4** (AIC-selected half-life, 0.832 model-based) to

0.870, a 3.8-point gain, the largest CR2 correction observed anywhere in this manuscript, enough for G9 to overtake every other specification tested, including G8 (0.863). This did not come with a Type I error cost (Section S11 of the Supplement checks this directly): every replicate converged under G9, and the power gain is at least consistent with CR2 correcting real conservatism rather than papering over a different problem. Full detail, the reference-cell table, and the heatmap are in the Supplement, Section S10.

3.7.9 S11: is G4's Type I error controlled?

G4 selects its half-life by AIC and then tests the interaction coefficient from the selected model, a post-selection inference problem. The same data drive both the selection and the test, which can in principle inflate Type I error beyond nominal. Block S11 checks this directly, refitting G1-G5 and G9 at the null ($c_{bm} = 0$) at the Hybrid reference cell, with $n_{sim} = 2000$ for tighter precision on a rate expected near 0.05.

The concern does not materialize. G4's Type I error is 0.038, conservative and in the same range as G1-G3 (0.034-0.036), and it is not inflated by the selection step. G9 (G4+CR2) is 0.052, indistinguishable from nominal 0.05. Read together with Section 3.6's power results, G9's status as the best-performing specification (Section 3.7, Block S10) is not an artifact of an uncontrolled test: it wins on power while holding Type I error at nominal, the same standard G3+CR2 (G8) is held to. This does not generalize beyond the single cell tested. Full detail and the heatmap are in the Supplement, Section S11.

3.8 A higher-precision check

Because a number of the Tier 1 gaps are small, we reran a focused 24-cell grid (OL+BDC design, $N \in \{35, 70\}$, true half-life $\in \{0.5, 1.0\}$) at 600 trials per cell, all converging. We compared specifications with a paired McNemar test, considerably more sensitive than an independent-groups comparison since all three are fit to the same trials. Type I error across the twelve null cells fell in $[0.003, 0.053]$, so nominal alpha holds. Paired McNemar tests place Exposure-weighted ahead of the unadjusted benchmark at all four alternative cells, by +2.3 to +13.7 percentage points. The largest gap is at the highest-leverage cell ($N = 70$, true $t_{1/2} = 1.0$ weeks, $p = 6.6 \times 10^{-17}$), with $p \leq 0.015$ at the remaining three, a sharper confirmation than the Tier 1 grid alone provides.

One scope caveat governs the rerun's third arm. It was produced by the package's `lme_analysis()` routine, whose carryover option appends time since discontinuation as a linear main effect rather than the lagged indicator L_{it} used in the Tier 1 grid. It is therefore not the Lag-adjusted specification of Section 2.4, and we do not report it as such here. That alternative remains a reasonable specification we have not yet evaluated (Section 4.6).

4 Discussion

4.1 The lagged-treatment term does no work

The clearest result of this study is a negative one. Adding the classical lagged-treatment indicator to the analysis model changed nothing measurable, anywhere in the grid. Across all 216 cells the mean power difference against the unadjusted benchmark was 0.003 against an MCSE of 0.018. Bias, MSE, and coverage agreed to three decimal places, and the equality held in every sensitivity block and all five cluster-robust stress cells under both inference procedures.

The strength of this conclusion rests on the design rather than the grid size. The two specifications estimate the same coefficient, were fitted to the same simulated datasets under common random numbers, and differ by exactly one term. There is therefore no confounding from a change of estimand, from Monte Carlo variation between arms, or from differential test size (both are conservative to the same degree, Section 3.6). What is left is the term itself, and, it turns out, it contributed nothing.

The mechanism follows from where the term sits. Carryover damages the interaction test on two fronts (Section 1). It attenuates the coefficient, because contaminated occasions are coded as drug-free, and it inflates the residual variance, because those occasions are heterogeneous in a way the model does not represent. The lagged indicator enters only as a nuisance main effect, with no $\text{bm}:L_{it}$ product term, so the tested interaction remains $\text{bm}:D_{it}$ and the attenuation is left untouched. It can in principle absorb some of the second effect, but it evidently absorbs too little to matter, since it marks a single occasion per randomization path and is constant thereafter, unable to describe a graded decline across a two- to four-occasion washout.

A more capable, elapsed-time-indexed nuisance term might succeed where the crude lagged indicator fails. We therefore ran a targeted check on the alternative the reference implementation provides, $\text{Sx} \sim \text{bm} + \text{t} + \text{Db} + \text{bm}:\text{Db} + \text{tsd}$, called the washout-adjusted specification (E4), across the three designs at true half-lives of 0.5 and 1.0 weeks ($N = 70$, $c_{bm} = 0.45$, exponential DGP, 500 replicates). It does slightly better, and the improvement is instructive rather than material. Type I error is at nominal for both (mean 0.050 versus 0.045). Power exceeds the benchmark in all six cells by a mean of 1.3 points (sign test $p = 0.031$), reaching significance under crossover at both half-lives (+1.8, $p = 0.008$; +2.2, $p = 0.003$) and OL+BDC at the longer one (+2.2, $p = 0.015$), but not Hybrid (+1.0, $p = 0.27$). The mechanism is exactly the one Section 2.4 predicts. Point estimates are unchanged, so nothing about the attenuation is repaired. The entire effect runs through the standard error, which only a long enough washout gives it something to describe. The crossover schedule places off-drug occasions 2.5 to 10 weeks after discontinuation, against one to two weeks for Hybrid. The practical ceiling on what any carryover nuisance term can deliver is therefore about two percentage points, and only in long-washout designs, against the nine to eleven points exposure weighting delivers where it works. This check covers only the exponential DGP at a single N and effect size, so it is best read as a probe rather than a fourth arm. It also shares every specification's structural limitation of entering carryover as a main effect only (Section

835 4.6). For practitioners the implication is direct. A carryover nuisance term of either form
836 should not be expected to recover meaningful power lost to carryover, and reporting one does
837 not constitute an effective adjustment for it.

838 4.2 The exposure-weighted advantage, and where it fails

839 The exposure-weighted predictor does help, but only conditionally. At its best cell (Hybrid,
840 $N = 70$, $c_{bm} = 0.45$, $t_{1/2} = 1.0$) it exceeds the unadjusted benchmark by about 10 points of
841 power (0.860 versus 0.766, roughly five MCSEs and so not attributable to chance), confirmed
842 at all four cells of the Section 3.8 high-precision rerun. Under the crossover design, though, it
843 performs considerably worse (0.488 versus 0.830). We think this happens because the within-
844 subject correlation there already carries most of the signal the decaying predictor was designed
845 to recover. It gains little in that setting, while its down-weighting of off-drug observations
846 works against it. The recommendation therefore holds only in the two non-crossover designs,
847 and should not be extended to a crossover trial, where the unadjusted binary indicator is
848 materially better.

849 That the one specification addressing attenuation is also the only one to gain power, while the
850 one addressing only residual variance gains nothing, suggests attenuation is the mechanism
851 that matters here. We read this as an interpretation consistent with the results, not as a
852 finding the design isolates directly.

853 4.3 What the robustness checks add

854 The Section 3.7 sensitivity blocks show the exposure-weighted advantage, where it exists, is
855 not uniformly fragile, but neither is it unconditionally robust. Under incorrect decay-shape
856 assumptions (Figure 6, Section 3.5) it widens on the light-tailed side and narrows to statistical
857 insignificance on the heavy-tailed side, disappearing entirely at the two most extreme Weibull
858 shapes examined ($k = 0.25, 0.5$). It survives an incorrectly assumed half-life, short of a large
859 over-assumption at a short true half-life (Block S2, Supplement Section S2), and it holds its
860 five- to eight-point lead across the autocorrelation range examined (Block S1, Supplement
861 Section S1). Decay-shape mis-specification is therefore the one robustness axis examined here
862 that can erode the advantage away entirely, rather than only narrowing it.

863 One caveat qualifies the Tier 1 figures throughout. Outside block S2, the analysis-side half-
864 life is set equal to the data-generating one. This grants Exposure-weighted an oracle no real
865 analyst possesses, so its Tier 1 margins should be read as upper bounds, and block S2 gives
866 the more realistic figure.

867 4.4 Dropout matters more than the method

868 The finding with the largest practical consequence concerns missing data, not carryover. When
869 patients drop out (Figure 7), power collapses for all three specifications together, to between
870 0.12 and 0.25 at 30% dropout, essentially indistinguishable. No specification offers protection,

871 since each can only organize whatever data remain. MAR dropout is somewhat worse than
872 MCAR, because the sickest patients, who carry the most signal, tend to leave first. Retaining
873 patients in the trial, in short, yields considerably more power than any choice among these
874 three specifications ever could.

875 4.5 What analysis is best in which situations, in practice

876 We collect the findings above into a decision rule rather than a set of pairwise comparisons:

- 877 • **Covariance moderation (Architecture B), Hybrid or OL+BDC design.** Use
878 Exposure-weighted with a CR2 cluster-robust standard error. The advantage holds
879 under half-life mis-specification short of a large over-assumption (Section 3.7, Block
880 S2), and across the autocorrelation range examined (Block S1). It holds and widens
881 under decay-shape mis-specification on the light-tailed side, but narrows to statistical
882 insignificance at the two most extreme heavy-tailed shapes examined (Section 3.5): this
883 recommendation should not be read as holding under an arbitrarily heavy-tailed decay
884 form. CR2 restores nominal Type I error without disturbing the ranking (Block S6,
885 validated at $N \in \{35, 70\}$ in the Hybrid design specifically; OL+BDC was not itself
886 stress-tested under CR2). This remains the recommended default: it has been through
887 the full robustness battery G4/G5 have not (next point).
- 888 • **G4 (AIC-selected half-life) and G5 (paired-difference) are promising at this**
889 **same cell, not yet validated defaults.** Both beat Exposure-weighted itself at the
890 Hybrid, Architecture B reference cell (Blocks S7-S8: G4 0.850, G5 0.842, G3 0.836, at
891 $t_{1/2} = 1.0$), and G4 in particular removes the need to assume a half-life at all. Neither
892 has been tested against decay-shape mis-specification, dropout, or under CR2 the way
893 G3 has, so this is grounds to investigate them further, not to displace the established
894 default yet. Adding a CR2 correction to G4 (G9 in the nine-way naming layer) goes
895 further still: at this same cell it outperforms every one of the other eight specifications,
896 including G3+CR2, with no accompanying Type I error inflation (Section 3.7, Blocks
897 S10-S11). This is the single best result anywhere in this manuscript, and also the least
898 validated. It rests on one cell, with none of G4's own robustness gaps yet closed.
- 899 • **Crossover (CO) design, any architecture.** Use Unadjusted. Exposure-weighted
900 is markedly worse under CO (Section 3.4), and Block S9 shows G4 and G5 fail here
901 too, G4 catastrophically. G4's AIC half-life selection systematically drifts to the largest
902 candidate in its grid under CO's sparse, widely spaced off-drug sampling (chosen in over
903 85% of replicates regardless of the true half-life), collapsing its power to roughly half of
904 Unadjusted's. It should specifically be avoided under any design with few, widely spaced
905 off-drug observations. G5 also underperforms Unadjusted under CO, though far more
906 mildly. The classical lagged-treatment remedy adds nothing here either (next point),
907 including in the one design where its washout is long enough to give it a chance (Section
908 4.1).
- 909 • **Architecture uncertain, or known to be mean moderation (Architecture A).**
910 Use Unadjusted or Lag-adjusted, not Exposure-weighted or G4. Both specifications'

911 advantage under covariance moderation reverses, marginally, under mean moderation
 912 (Block S8). **G5** is the one specification whose performance does not depend much on
 913 which architecture generated the data, small under B, smaller under A, but it has been
 914 tested at only one design and reference cell outside the Hybrid comparison (Block S9).
 915 • **Never rely on Lag-adjusted as a carryover remedy.** It is statistically indistin-
 916 guishable from Unadjusted everywhere tested in this manuscript, Tier 1 and every Tier
 917 2 block alike (Section 4.1). If a reviewer or protocol expects some carryover term to be
 918 present, including it costs nothing measurable, but it should not be reported as having
 919 adjusted for carryover.
 920 • **Minimize dropout before optimizing specification choice.** It is the single largest
 921 lever in the entire grid (Section 4.4), and no specification, including **G4/G5**, has been
 922 shown to offer any protection against it; their behavior under dropout is untested.

923 4.5.1 Why **G4** and **G5** succeed at Hybrid but fail under CO

924 Both new specifications' Hybrid, Architecture B advantage evaporates under the CO design
 925 (Block S9), for two different, design-specific reasons rather than a shared structural flaw of the
 926 kind that sank **E5/E6** (Section 4.6). **G4**'s failure is mechanical. With only a few, widely spaced
 927 off-drug occasions, the likelihood surface across candidate half-lives is weakly informative, and
 928 AIC selection drifts to the boundary of its candidate grid rather than locating the true value,
 929 systematically overestimating the half-life. Section 3.7's S2 block already showed that over-
 930 assumption is the more damaging direction of half-life error when the half-life is fixed by
 931 the analyst. Under CO, **G4** reproduces that same damaging error through the estimation
 932 procedure itself, on nearly every replicate.

933 This yields a concrete, testable prediction. **G4** should be reliable whenever a design gives
 934 several off-drug occasions across a graded range of elapsed time, as Hybrid does despite its
 935 short overall washout, and unreliable whenever off-drug sampling is sparse, however long the
 936 washout runs in calendar time.

937 **G5**'s failure has no comparable mechanical account and is milder. The most plausible explana-
 938 tion is that CO's own within-subject contrast (Section 3.4) already carries most of the signal
 939 a between-patient paired-difference regression is built to extract, so discarding the repeated-
 940 measures and correlation structure costs more under CO than it does under Hybrid, where
 941 that structure is doing more work. This was not tested directly, however, and would require
 942 comparing **G5** against the within-cell decomposition of **G3**'s advantage to confirm.

943 Two implications follow. First, these results reinforce rather than undercut the earlier **E5/E6**
 944 diagnosis (Section 4.6). The more fundamental unresolved question is not which design an
 945 alternative specification is tested under, but whether the underlying data-generating process
 946 rewards that specification's added structure at all, and neither **G4** nor **G5** was tested against
 947 a DGP built to favor them specifically. Second, the practical recommendation is narrower
 948 than 'estimate the half-life' or 'use a simpler method': both **G4** and **G5** are candidates worth
 949 validating further within the Hybrid/OL+BDC, Architecture B cell where this manuscript's

950 other recommendations already concentrate, not general-purpose replacements for Exposure-
951 weighted or Unadjusted.

952 4.6 Scope, related work, and limits

953 This manuscript deliberately narrows the full-scope grid to the covariance-moderation
954 architecture and to the non-linear decay forms, in the interest of a self-contained eval-
955 uation of manageable length. The archived full-scope drafts (`archive/report.Rmd` and
956 `archive/report-slim.Rmd`) show that the exposure-weighted advantage documented here
957 does not carry over to mean moderation, where the biomarker moderates the mean response
958 directly and Exposure-weighted is marginally dominated in every design examined. Readers
959 who need that cross-architecture comparison should consult those drafts rather than extrap-
960 olate from this one. Establishing what the originally published power values (Section 3.3)
961 were computed under remains outstanding work.

962 The crossover tradition surveyed by Jones and Kenward⁴ would lead one to expect a shape-
963 free lagged term to outperform a parametric predictor once the decay shape is mis-specified.
964 Our results do not support that expectation over the range examined, for two reasons. The
965 exposure-weighted penalty for an incorrect half-life is too small to be overtaken (Section
966 3.7). And more fundamentally, the lagged term does not improve on ignoring carryover at all
967 (Section 4.1), leaving no advantage for decay-shape mis-specification to erode.

968 One adjacent specification bounds the negative result of Section 4.1. Every carryover term
969 considered there, lagged and washout-adjusted alike, enters as a nuisance main effect with
970 no product against the biomarker. That finding is therefore that a carryover *main effect*
971 does not recover the attenuated signal, not that no shape-free specification can. The natural
972 candidate is instead a carryover term interacted with the biomarker, either $Sx \sim bm + t + Db$
973 $+ bm:Db + L + bm:L$ or its washout-adjusted counterpart with `bm:tsd`, letting the biomarker
974 effect decline across the washout without committing to a rate.

975 An earlier version of Block S7 tested exactly these two specifications and found a clean neg-
976 ative result at every cell examined. We traced this to a structural cause: this manuscript's
977 DGP has a single true decay parameter, so a second, interacted carryover coefficient adds
978 estimation noise without a corresponding signal to recover. They were replaced by two spec-
979 ifications addressing the same open problem, the analyst's unknown decay half-life, from
980 different angles: **G4** (AIC-selected half-life) and **G5** (a paired-difference regression with no
981 carryover representation at all).

982 Both beat Unadjusted at the Hybrid, Architecture B reference cell, **G4** even edging out
983 Exposure-weighted itself (Block S7). Neither survives a change of trial design or architecture
984 (Blocks S8-S9). **G4**'s AIC selection fails specifically under sparse off-drug sampling (Section
985 4.5), and **G5**'s advantage appears to depend on the repeated-measures structure it discards
986 being otherwise underused, which is itself design-dependent. Block S8 shows the Lag-adjusted
987 null of Section 3.4 replicates under Architecture A, so it is not an artifact specific to covari-
988 ance moderation. Exposure-weighted's advantage, and now **G4**'s, are architecture-sensitive,
989 reversing in direction, though not substantially, under Architecture A (Section 3.7, Block S8).

Three further N-of-1 analysis approaches sit outside the G1-G9 family and are not evaluated here, because each requires substantial independent methodological development rather than a drop-in specification within our existing pipeline.

Tang and Landes¹⁰ propose formula-based t -tests that correct a single-subject paired-difference test's standard error and degrees of freedom for an estimated first-order serial correlation, the natural refinement G5 lacks (G5 discards within-patient correlation entirely). Their tests have no biomarker-moderation term, though, and would need a `diff ~ bm` extension with a serial-correlation-aware variance to enter our comparison.

Schmid and Yang¹¹ describe a Bayesian hierarchical alternative to the frequentist mixed-model and sandwich-estimator family spanning G1-G9, with partial pooling of individual-level treatment-effect, trend, and autocorrelation parameters across patients. They note that carryover is tractable only in aggregated, multi-individual analyses like ours, but a biomarker-moderation extension of their model is a separate undertaking.

Daza¹² frames N-of-1 inference as a counterfactual, time-varying-treatment problem, estimated via g-computation or inverse-probability weighting. This is a structurally different paradigm from G1-G9's explicit exponential-decay or lag covariates: carryover there is confounding to be controlled rather than a predictor to be modeled.

None of the three is a candidate for a tenth specification within this manuscript's scope, but each identifies a direction future work on this estimand should consider.

Two further limitations bound what we can conclude. The numbers presented here come from a single parameter set calibrated to a prazosin and PTSD trajectory, so absolute power values should be read as illustrative, and the ranking, itself conditional on trial design, as the more portable result. And the sensitivity checks vary one factor at a time, so they can miss genuine interactions between factors, for instance between high autocorrelation and heavy dropout. The one joint check available to us, block S5 in the full-scope manuscript, revealed no such interaction.

5 Conclusions

1. **The lagged-treatment term does nothing.** It never improved on ignoring carryover altogether, on any measure, in any design, at any half-life, and in every sensitivity block, a clean null given the two specifications share an estimand and differ by a single term. The term enters as a nuisance main effect only, leaving the attenuation of the interaction untouched, and should not be reported as an adjustment for carryover.
2. **Exposure weighting helps, but conditionally.** It achieves the highest power, about 10 points over the unadjusted benchmark, only in the Hybrid and OL+BDC designs. Its lead survives mis-specified half-life there, and widens under a light-tailed decay-shape mis-specification, but narrows to statistical insignificance at the two most heavy-tailed Weibull shapes examined; it is not robust to an arbitrarily heavy decay tail. Under crossover it is markedly worse than the binary indicator (0.488 against 0.830) and should

not be used. Where used, we recommend a CR2 cluster-robust standard error (Block S6).

3. **An approximate half-life is adequate, but the analyst should err short.** A fourfold error at a true half-life of one week costs 6.6 points of power, so a rough pharmacokinetic estimate suffices in practice. The error is asymmetric. Under-assuming drives the predictor toward the still-serviceable binary indicator, while over-assuming erodes the advantage to a tie. Outside block S2 the analyst is granted the true half-life, so the Tier 1 margins should be read as upper bounds.

4. **Dropout matters more than the choice of method.** Power falls sharply as dropout increases, roughly halving at 20%, equally across all three specifications. Preventing dropout is a more effective use of design effort than any choice among them.

5. **The interaction test is somewhat conservative, and correctly so.** All three specifications reject slightly less often than nominal under the null. A CR2 cluster-robust standard error restores the correct rate while recovering modest power, and the exposure-weighted advantage is held or widened under CR2 at four of five S6 stress cells (Block S6).

6. **This is a narrowed slice of a larger study.** Mean moderation and linear decay fall outside this manuscript's scope. The archived drafts `archive/report.Rmd` and `archive/report-slim.Rmd` are the authority on those comparisons and should be consulted before generalizing beyond covariance moderation. We also make no claim of numerical agreement with the originally published power values (Section 3.3).

7. **Carryover terms interacted with the biomarker do not help; two alternative-paradigm specifications do, but only at the reference cell.** Crossing a lagged or washout carryover term with the biomarker gave lower power than the unadjusted benchmark at every cell examined (Block S7's original version). We traced this to a structural cause: this manuscript's DGP has a single true decay parameter, so a second, interacted coefficient adds estimation noise without a corresponding signal to recover (item 1's mechanism, restated). They were replaced by two specifications answering the same open question, the analyst's unknown decay half-life, differently. **G4** estimates it by AIC selection; **G5** discards the decay representation, and the mixed model itself, for a two-stage paired-difference regression following Senn, Julious and Araujo⁷. Both beat Exposure-weighted at the Hybrid, Architecture B reference cell (Blocks S7-S8; **G4** 0.850, **G5** 0.842, **G3** 0.836 at $t_{1/2} = 1.0$). Neither result generalizes: Block S8 shows both reverse, like Exposure-weighted itself, under Architecture A, and Block S9 shows both fail under the crossover design, **G5** mildly and **G4** catastrophically. **G4**'s CO failure has a clear mechanism: AIC selection systematically drifts to the boundary of its candidate half-life grid when off-drug sampling is sparse and widely spaced, reproducing the S2 finding that over-assuming the half-life is the more damaging error, now through the estimation procedure itself rather than a fixed wrong assumption. Both specifications should be read as promising candidates for further validation at the one cell type

where this manuscript already recommends Exposure-weighted, not as general-purpose replacements for it or for Unadjusted.

8. **Adding CR2 to the AIC-selected specification gives the single best result in this manuscript, at one cell only.** G9 (G4 plus a CR2 cluster-robust standard error) outperforms all eight other specifications at the Hybrid, Architecture B reference cell, including G3+CR2, and its Type I error holds at nominal despite the post-selection risk G4's half-life selection step raises in principle (Section 3.7, Blocks S10-S11). We read this as a promising direction, not a recommendation: G9 has not been tested against decay-shape mis-specification, dropout, a crossover design, or an alternative architecture, all of which item 2's specification (G3+CR2) has already been checked against.

6 Conflict of interest

The authors declare no conflicts of interest.

7 Funding

This work received no specific funding.

8 Data availability statement

The data and code supporting the findings of this study are available in the pmsimstats-ng project repository. Per-replicate Monte Carlo outputs and ADEMP-compliant cell-level summaries are versioned under `analysis/data/`. Simulation drivers are at `analysis/scripts/carryover-sensitivity/`. Exact paths for this manuscript are listed in the Reproducibility section below.

9 Reproducibility

This manuscript was rendered with R 4.6.0 inside the project's zzzcollab Docker container. The production simulation ran under R 4.5.3 (§3.6), with the image hash in `Dockerfile` and the package set in `renv.lock`. The simulation study uses `nlme` for the continuous-time linear-mixed analysis, `parallel` for the 8-core parameter-grid sweep, and `data.table` or `furrr/purrr` in the two simulation collections. No simulation code runs during knitting; knitting loads pre-computed checkpoints only. Each driver sets a master seed (20260507) and derives per-replicate seeds from it via `parallel::mc.set.seed`.

Data and code availability. All simulation outputs, cell-level summaries, and driver scripts are versioned under `analysis/scripts/carryover-sensitivity/` and `analysis/data/` in the pmsimstats-ng repository. This includes the principal factorial, the S1 to S5 sensitivity

blocks, the S6 cluster-robust recalibration, the washout-adjusted (E4) check of Section 4.1, and
 the high-precision prototype rerun of Section 3.7. It also includes the S7 alternative-paradigm
 block (`09-run-sensitivity-s7.R`, `output/09-sensitivity-s7.rds`); the S8 architecture-
 by-half-life block (`10-run-sensitivity-s8.R`, `output/10-sensitivity-s8.rds`); the S9
 CO-design block (`14-run-sensitivity-s9.R`, `output/14-sensitivity-s9.rds`); the S10
 CR2 comparison block (`15-run-sensitivity-s10.R`, `output/15-sensitivity-s10-g.rds`);
 and the S11 Type I error block (`16-run-sensitivity-s11.R`, `output/16-sensitivity-s11.rds`).
 All five were added 2026-08-18/19 (master seed 20260415, 500 replicates per cell except
 S11's 2000; Hybrid reference cell except S9, which uses CO, and S11, which sets $c_{bm} = 0$).
 E5/E6 (biomarker-interacted carryover terms) were tested in an earlier version of S7-S9 and
 are documented in Section 4.6; they were replaced by G4 (AIC-selected half-life, reusing
 the candidate-grid logic of `characterize_carryover()`, `R/carryover_analysis.R`) and G5
 (paired-difference regression). The E1/E3/E2, E7, E9 fitters, and the E1cr2/E3cr2/E7cr2
 CR2 variants used by S10 (`cr2_extract()`, `fit_spec_s7`, `simulate_cell_s7`), are ap-
 pended to `simulation-core.R` alongside the Tier 1 `fit_spec/fit_all_specs` pair, which
 they do not modify. `spec-labels.R` additionally defines the G1-G9 compact display-code
 layer (`g_code`, `g_labels`) used in S10's table and figure; G8 (G3+CR2, stored as E2cr2)
 is reused directly from Block S6 rather than rerun, so it does not share common random
 numbers with the rest of S10's comparison. Figure 4, and the decay-form mismatch panel
 (Figure 6), retain the original three specifications (G1-G3), since decay shape is the only
 axis those two figures vary. Both were regenerated from a dedicated 30-cell simulation
 at a more extreme Weibull shape range ($k = 0.25, 0.5, 2.0, 4.0$, extended from an initial
 $k = 0.5, 2.0$, itself replacing an earlier $k = 0.7, 1.5$; $k = 1.0$ is not a separate condition, since
 it duplicates Exponential exactly) than the shared production grid (`02-grid-summary.rds`)
 still uses (`23-run-decay-shape-sensitivity.R`, `02-decay-shape-sensitivity.rds`,
 $n_{sim} = 500$, Architecture B only). This is a standalone script rather than a change to
`01-run-factorial.R`'s shared grid, since that grid is also consumed by the archived full-
 scope drafts and other manuscripts. Figures 2 and 3 were later expanded to all nine G1-G9
 specifications, from a dedicated 30-cell simulation restricted to the exponential DGP and Ar-
 chitecture B (`19-run-tier1-hendrickson-g9.R`, `02-grid-summary-hendrickson-g9.rds`,
 $n_{sim} = 500$). This replaced the original three-specification version of both panels rather
 than supplementing it, and superseded the separate Type I error checks the manuscript
 previously reported at Figures 6-9 (the Tier 1 Type I boxplot/heatmap and the S6 G1-
 G3/G1-G9 Type I dot plot/heatmaps), since the $c_{bm} = 0$ column of Panel A now reports
 Type I error for all nine specifications directly. Those retired figures' underlying data
 and scripts (`03-render-figures-extra-slim.R`, `07-type1-cr2-figure-extra-slim.R`,
`18-run-sensitivity-s6-g9.R`) are retained for archival purposes but no longer referenced
 by this document. The corresponding Tier 2 heatmaps (S1-S4, S6, S7, S8, S9, S10, S11)
 are produced by `13-hendrickson-heatmap-tier2.R` from `02-sensitivity-summary.rds`
 and the S7/S8/S9/S10/S11 output `.rds` files (plus `diag-s6-cr2.rds` for G8 and Table 4);
 these Tier 2 figures are additional to, not replacements for, the existing S1-S4 line plots
 and the S7/S8 kable tables. The grid design and Morris ADEMP pre-registration are
 in `simulation-grid-plan.md` in the same directory. Superseded wider-scope drafts are

retained under `archive/`, documented in its `README.md`.

Specification labels. All nine specifications are stored under E-prefixed codes (E1, E2, E3, E7, E9, E1cr2, E2cr2, E3cr2, E7cr2) in every data file and script, and this manuscript displays them uniformly as G1-G9, or by name, at render time only. The mapping is applied by `spec-labels.R`, so the stored values and archived outputs are unaffected. What a reader sees as G3 (Exposure-weighted) in the prose above is stored as E2 in every `.rds` file this manuscript reads. Output archived before 2026-07-31 stores the specifications as A1, A2, A3; the migration is documented in `analysis/report/NOTATION.md`.

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